EXAMPLE 1 | Determine the equilibrium of the difference equation $N_{t+1} = RN_t$, where $R > 0$, and classify it as stable or unstable.

SOLUTION The equilibrium $\hat{N}$ satisfies the equation $\hat{N} = R\hat{N}$. The only solution of $(R - 1)\hat{N} = 0$ is $\hat{N} = 0$, unless $R = 1$. We know that the solution of the recursion $N_{t+1} = RN_t$ is $N_t = N_0 \cdot R^t$. There are three cases:

- If $0 < R < 1$, then $N_t = N_0 \cdot R^t \to 0$ as $t \to \infty$, so $N_t \to \hat{N} = 0$. Therefore the equilibrium $\hat{N} = 0$ is stable in this case.
- If $R > 1$, then $N_t = N_0 \cdot R^t \to \infty$ as $t \to \infty$, and so the equilibrium $\hat{N} = 0$ is unstable in this case.
- If $R = 1$, then $N_t = N_0$ for all t. This case is called *neutral*. ■

Cobwebbing

There is a graphical method for finding equilibria and determining whether they are stable or unstable. It is called **cobwebbing** and is illustrated in Figure 2. We start with an initial value x_0 on the horizontal axis and locate $x_1 = f(x_0)$ as the distance from the point x_0 up to the point (x_0, x_1) on the graph of f. Then we draw the horizontal line segment from (x_0, x_1) to the point (x_1, x_1) on the diagonal line. The point x_1 lies directly beneath (x_1, x_1) on the horizontal axis.

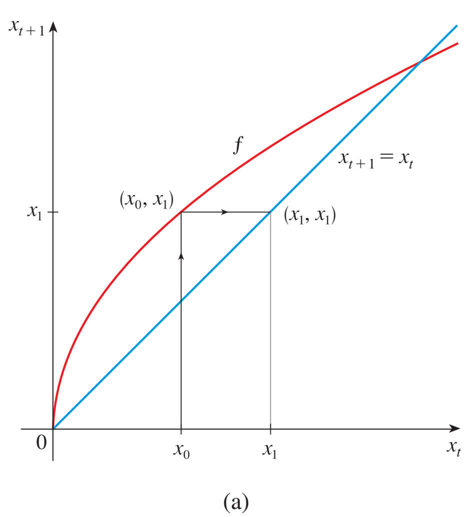


FIGURE 2
Cobwebbing

In Figure 2(b) we repeat this procedure to obtain x_2 from x_1, drawing a vertical line segment from (x_1, x_1) to (x_1, x_2) on the graph of f and then a horizontal line segment over to the diagonal. Continuing in this manner we create a zigzag path that reflects off the diagonal line and shows how the successive terms in the sequence can be obtained geometrically.

EXAMPLE 2 | Use cobwebbing to determine whether the equilibria $\hat{x} = 0$, $\hat{x} = a$, and $\hat{x} = b$ in Figure 1 are stable or unstable.

SOLUTION Figure 3 is a larger version of Figure 1. We experiment with different initial values and use cobwebbing to visualize the values of x_t. We notice that

$$\text{if } a < x_0 < b, \text{ then } \lim_{t \to \infty} x_t = b$$

$$\text{but } \quad \text{if } 0 < x_0 < a, \text{ then } \lim_{t \to \infty} x_t = 0$$

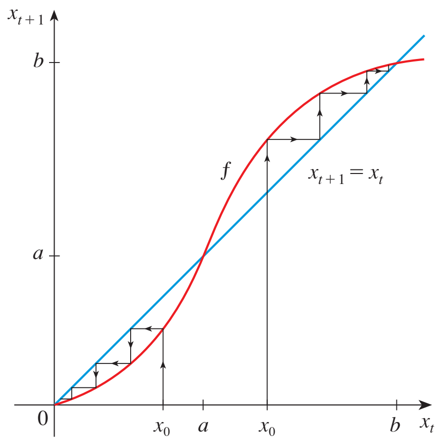


FIGURE 3
Cobwebbing with stable and unstable equilibria

Solutions that start close to b approach b, so $\hat{x} = b$ is a stable equilibrium. Likewise, solutions that start close to 0 approach 0, so $\hat{x} = 0$ is also a stable equilibrium. But solutions that start close to a (on either side of a) move away from a. So $\hat{x} = a$ is an unstable equilibrium. ■

So far we have used cobwebbing only with increasing functions f. Figure 4 shows what happens when f decreases. We apply cobwebbing with initial value x_0 to a difference equation $x_{t+1} = f(x_t)$ with decreasing f. Instead of the zigzag paths in Figures 2 and 3, you can see that we get spiral paths and the values of x_t oscillate around the equilibrium $\hat{x}$. In Figure 4(a), $x_t \to \hat{x}$ as $t \to \infty$, so $\hat{x}$ is stable. In Figure 4(b), however, the values of x_t move away from $\hat{x}$, so $\hat{x}$ is unstable.

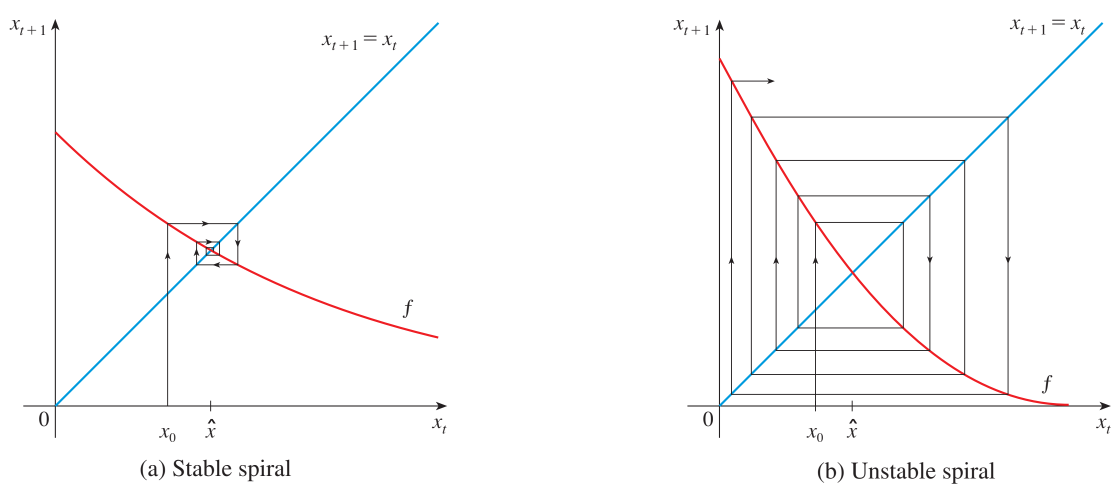


FIGURE 4 Spiral cobwebbing

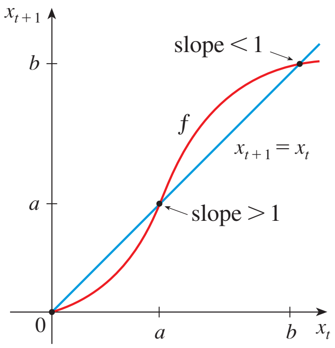


FIGURE 5

Stability Criterion

An equilibrium occurs when the graph of f crosses the diagonal line, which has slope 1. Figure 5 shows the increasing function f from Figure 3 and we see that at the stable equilibrium $\hat{x} = b$ the curve crosses the diagonal from above to below, so $f'(b) < 1$. At the unstable equilibrium $\hat{x} = a$ the curve crosses the diagonal from below to above, so $f'(a) > 1$.

If f is decreasing, we see from diagrams like Figure 4 that stable spirals occur when $-1 < f'(\hat{x}) < 0$ and unstable spirals occur for steeper curves, that is, $f'(\hat{x}) < -1$.

To summarize, our intuition tells us that equilibria are stable when $-1 < f'(\hat{x}) < 1$ and unstable when $f'(\hat{x}) > 1$ or $f'(\hat{x}) < -1$. So the following theorem appears plausible. A proof, using the Mean Value Theorem, appears in Appendix E.

(3) The Stability Criterion for Recursive Sequences Suppose that $\hat{x}$ is an equilibrium of the recursive sequence $x_{t+1} = f(x_t)$, where f' is continuous. If $|f'(\hat{x})| < 1$, the equilibrium is stable. If $|f'(\hat{x})| > 1$, the equilibrium is unstable.

Let's revisit some of the difference equations we studied in Section 2.1 and see how the Stability Criterion applies to those equations.

EXAMPLE 3 | **BB** Drug concentration In Example 2.1.5 we considered the difference equation

$$C_{n+1} = 0.3C_n + 0.2$$

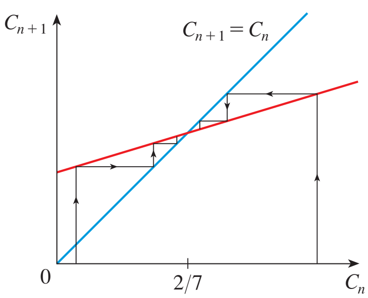


FIGURE 6

where C_n is the concentration of a drug in the bloodstream of a patient after injection on the nth day, 30% of the drug remains in the bloodstream the next day, and the daily dose raises the concentration by 0.2 mg/mL.

Here the recursion is of the form $C_{n+1} = f(C_n)$, where $f(x) = 0.3x + 0.2$. The equilibrium concentration is $\hat{C}$, where $0.3\hat{C} + 0.2 = \hat{C}$. Solving this equation gives $\hat{C} = \frac{2}{7}$. The derivative of f is $f'(\hat{C}) = 0.3$, which is less than 1, so the equilibrium is stable, as illustrated by the cobwebbing in Figure 6. In fact, in Section 2.1 we calculated that

$$\lim_{n\to\infty} C_n = \frac{2}{7}$$ ■

EXAMPLE 4 | **BB** Logistic difference equation In Example 2.1.8 we examined the long-term behavior of the terms defined by the logistic difference equation

$$x_{t+1} = cx_t(1 - x_t)$$

for different positive values of c. Use the Stability Criterion to explain that behavior.

SOLUTION We can write the logistic equation as $x_{t+1} = f(x_t)$, where

$$f(x) = cx(1 - x)$$

We first find the equilibria by solving the equation $f(x) = x$:

$$cx(1 - x) = x \iff x = 0 \quad \text{or} \quad c(1 - x) = 1$$

So one equilibrium is $\hat{x} = 0$. To find the other one, note that

$$c - cx = 1 \iff c - 1 = cx \iff x = \frac{c - 1}{c} = 1 - \frac{1}{c}$$

So the other equilibrium is

$$\hat{x} = 1 - \frac{1}{c}$$

The derivative of $f(x) = c(x - x^2)$ is $f'(x) = c(1 - 2x)$. For the first equilibrium, $\hat{x} = 0$, we have $f'(0) = c$, so the Stability Criterion tells us that $\hat{x} = 0$ is stable if $0 < c < 1$ and unstable if $c > 1$.

For the second equilibrium, $\hat{x} = 1 - 1/c$, we have

$$f'\left(1 - \frac{1}{c}\right) = c\left[1 - 2\left(1 - \frac{1}{c}\right)\right] = c\left(\frac{2}{c} - 1\right) = 2 - c$$

The Stability Criterion says that this equilibrium is stable if $|2 - c| < 1$. But

$$|2 - c| < 1 \iff -1 < 2 - c < 1 \iff 1 < c < 3$$

We also note that $f'(\hat{x})$ is negative when $2 - c < 0$, that is, $c > 2$, so oscillation occurs when $c > 2$. Let's compile all this information in the following chart:

	$\hat{x} = 0$	$\hat{x} = 1 - \frac{1}{c}$
$0 < c < 1$	stable	
$1 < c < 2$	unstable	stable
$2 < c < 3$	unstable	stable (oscillation)
$c > 3$	unstable	unstable (oscillation)

In Exercises 17–20 you are asked to illustrate the four cases in the chart in Example 4 both by cobwebbing and by graphing the recursive sequence.

Referring to the chart, we find an explanation for what we noticed in Example 2.1.8. When $c < 3$, one of the equilibria is stable and so the terms converge to that number. But when $c > 3$ both equilibria are unstable and so the terms have nowhere to go; they don't approach any fixed number. ■

EXAMPLE 5 | BB Ricker equation W. E. Ricker introduced the discrete-time model

$$x_{t+1} = cx_t e^{-x_t} \qquad c > 0$$

in the context of modeling fishery populations. Find the equilibria and determine the values of c for which they are stable.

SOLUTION The Ricker equation is $x_{t+1} = f(x_t)$, where

$$f(x) = cxe^{-x}$$

To find the equilibria we solve the equation $f(x) = x$:

$$cxe^{-x} = x \iff x = 0 \text{ or } ce^{-x} = 1$$

One equilibrium is $\hat{x} = 0$. The other satisfies

$$ce^{-x} = 1 \iff c = e^x \iff x = \ln c$$

The second equilibrium is $\hat{x} = \ln c$.

We use the Product Rule to differentiate f:

$$f'(x) = cx(-e^{-x}) + ce^{-x} = c(1 - x)e^{-x}$$

For $\hat{x} = 0$ we have $f'(0) = c$, so it is stable if $0 < c < 1$ and unstable if $c > 1$. For $\hat{x} = \ln c$ we get

$$f'(\ln c) = c(1 - \ln c)e^{-\ln c} = c(1 - \ln c) \cdot \frac{1}{c} = 1 - \ln c$$

Therefore

$$|f'(\hat{x})| < 1 \iff |1 - \ln c| < 1 \iff -1 < 1 - \ln c < 1$$

Now

$$1 - \ln c < 1 \iff \ln c > 0 \iff c > 1$$

and

$$-1 < 1 - \ln c \iff \ln c < 2 \iff c < e^2$$

By the Stability Criterion, $\hat{x} = \ln c$ is stable when

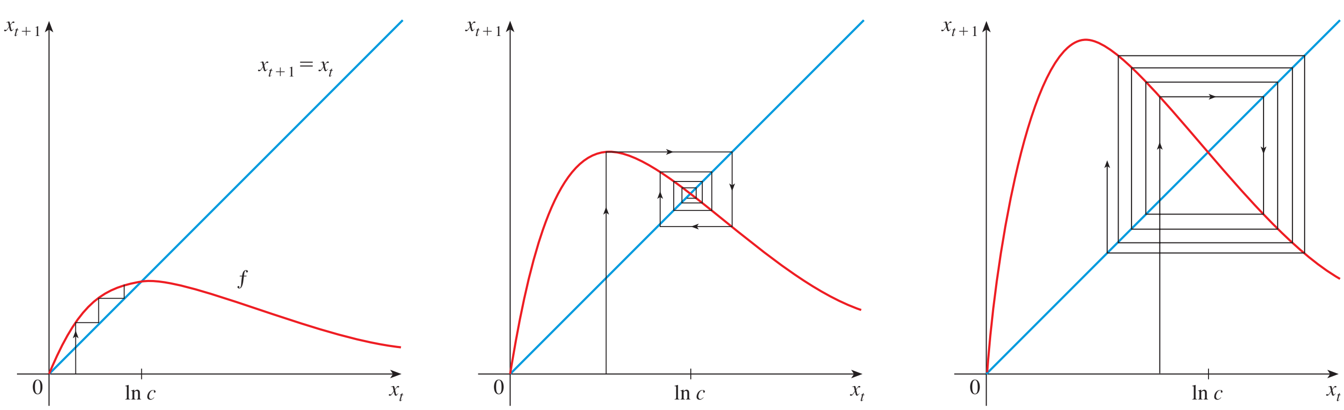
$$1 < c < e^2$$

When $0 < c < 1$ or $c > e^2$, $\hat{x} = \ln c$ is unstable.

We also note that oscillation occurs when $f'(\hat{x}) < 0$, so

$$1 - \ln c < 0 \Rightarrow \ln c > 1 \Rightarrow c > e$$

Figure 7 illustrates cobwebbing for the Ricker equation for three values of c.



$c = 2.5$ stable

$c = 6$ stable spiral

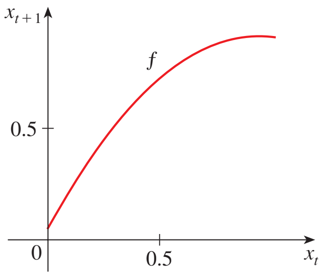
$c = 9$ unstable spiral

FIGURE 7

EXERCISES 4.5

1–4 The graph of the function f for a recursive sequence $x_{t+1} = f(x_t)$ is shown. Estimate the equilibria and classify them as stable or unstable. Confirm your answer by cobwebbing.

1.



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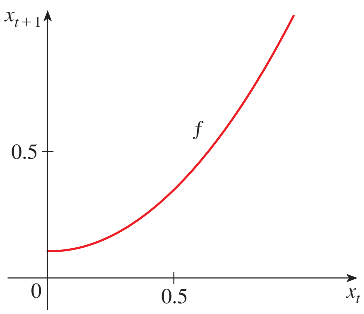

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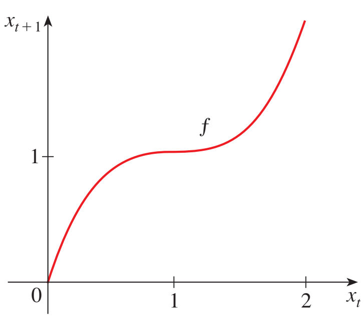
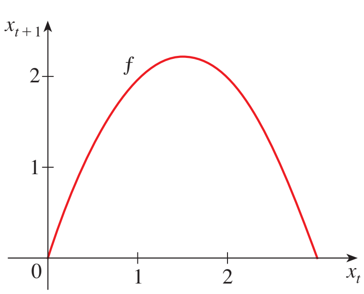


4.



5–10 Find the equilibria of the difference equation and classify them as stable or unstable.

5. $x_{t+1} = \frac{1}{2}x_t^2$

6. $x_{t+1} = 1 - x_t^2$

7. $x_{t+1} = \dfrac{x_t}{0.2 + x_t}$

8. $x_{t+1} = \dfrac{3x_t}{1 + x_t}$

9. $x_{t+1} = 10x_t e^{-2x_t}$

10. $x_{t+1} = x_t^3 - 3x_t^2 + 3x_t$

11–12 Find the equilibria of the difference equation and classify them as stable or unstable. Use cobwebbing to find $\lim_{t\to\infty} x_t$ for the given initial values.

11. $x_{t+1} = \dfrac{4x_t^2}{x_t^2 + 3}, \quad x_0 = 0.5, \quad x_0 = 2$

12. $x_{t+1} = \dfrac{7x_t^2}{x_t^2 + 10}, \quad x_0 = 1, \quad x_0 = 3$

13–14 Find the equilibria of the difference equation. Determine the values of c for which each equilibrium is stable.

13. $x_{t+1} = \dfrac{cx_t}{1 + x_t}$

14. $x_{t+1} = \dfrac{x_t}{c + x_t}$

15. Drug pharmacokinetics A patient takes 200 mg of a drug at the same time every day. Just before each tablet is taken, 10% of the drug remains in the body.

(a) If Q_n is the quantity of the drug in the body just after the nth tablet is taken, write a difference equation expressing Q_{n+1} in terms of Q_n.

(b) Find the equilibria of the equation in part (a).

(c) Draw a cobwebbing diagram for the equation.

16. Drug pharmacokinetics A patient is injected with a drug every 8 hours. Immediately before each injection the concentration of the drug has been reduced by 40% and the new dose increases the concentration by 1.2 mg/mL.

(a) If Q_n is the concentration of the drug in the body just after the nth injection is given, write a difference equation expressing Q_{n+1} in terms of Q_n.

(b) Find the equilibria of the equation in part (a).

(c) Draw a cobwebbing diagram for the equation.

17–20 Logistic difference equation Illustrate the results of Example 4 for the logistic difference equation by cobwebbing and by graphing the first ten terms of the sequence for the given values of c and x_0.

17. $c = 0.8, \quad x_0 = 0.6$

18. $c = 1.8, \quad x_0 = 0.1$

19. $c = 2.7, \quad x_0 = 0.1$

20. $c = 3.6, \quad x_0 = 0.4$

21. Sustainable harvesting In Example 4.4.5 we looked at a model of sustainable harvesting, which can be formulated as a discrete-time model:

$$N_{t+1} = N_t + rN_t\left(1 - \frac{N_t}{K}\right) - hN_t$$

Find the equilibria and determine when each is stable.

22. Heart excitation A simple model for the time x_t it takes for an electrical impulse in the heart to travel through the atrioventricular node of the heart is

$$x_{t+1} = \frac{375}{x_t - 90} + 100 \qquad x_t > 90$$

(a) Find the relevant equilibrium and determine when it is stable.
(b) Draw a cobwebbing diagram.

Source: Adapted from D. Kaplan et al., *Understanding Nonlinear Dynamics* (New York: Springer-Verlag, 1995).

23. Species discovery curves A common assumption is that the rate of discovery of new species is proportional to the fraction of currently undiscovered species. If d_t is the fraction of species discovered by time t, a recursion equation describing this process is

$$d_{t+1} = d_t + a(1 - d_t)$$

where a is a constant representing the discovery rate and satisfies $0 < a < 1$. Find the equilibria and determine the stability.

24. Drug resistance in malaria In the project on page 78 we developed the following recursion equation for the spread of a gene for drug resistance in malaria:

$$p_{t+1} = \frac{p_t^2 W_{RR} + p_t(1 - p_t)W_{RS}}{p_t^2 W_{RR} + 2p_t(1 - p_t)W_{RS} + (1 - p_t)^2 W_{SS}}$$

where W_{RR}, W_{RS}, and W_{SS} are constants representing the probability of survival of the three genotypes. In fact this model applies to the evolutionary dynamics of any gene in a population of diploid individuals.
(a) Find the equilibria of the model in terms of the constants.
(b) Suppose that $W_{RR} = \frac{3}{4}$, $W_{RS} = \frac{1}{2}$, and $W_{SS} = \frac{1}{4}$. Determine the stability of each equilibrium (provided it lies between 0 and 1). Plot the cobwebbing diagram and interpret your results.
(c) Suppose that $W_{RR} = \frac{1}{2}$, $W_{RS} = \frac{3}{4}$, and $W_{SS} = \frac{1}{4}$. Determine the stability of each equilibrium. Plot the cobwebbing diagram and interpret your results.

25. Blood cell production A simple model of blood cell production is given by

$$R_{t+1} = R_t(1 - d) + F(R_t)$$

where d is the fraction of red blood cells that die from one day to the next and $F(x)$ is a function specifying the number of new cells produced in a day, given that the current number is x. Find the equilibria and determine the stability in each case.
(a) $F(x) = \theta(K - x)$, where θ and K are positive constants
(b) $F(x) = \dfrac{ax}{b + x^2}$, where a and b are positive constants and $a > bd$

Source: Adapted from N. Mideo et al., "Understanding and Predicting Strain-Specific Patterns of Pathogenesis in the Rodent Malaria *Plasmodium chabaudi*," *The American Naturalist* 172 (2008): E214–E328.

4.6 Antiderivatives

Suppose you know the rate at which a bacteria population is increasing and want to know the size of the population at some future time. Or suppose you know the rate of decrease of your blood alcohol concentration and want to know your BAC an hour from now. In each case, the problem is to find a function F whose derivative is a known function f. If such a function F exists, it is called an *antiderivative* of f.

> **Definition** A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x in I.

For instance, let $f(x) = x^2$. It isn't difficult to discover an antiderivative of f if we keep the Power Rule in mind. In fact, if $F(x) = \frac{1}{3}x^3$, then $F'(x) = x^2 = f(x)$. But the function $G(x) = \frac{1}{3}x^3 + 100$ also satisfies $G'(x) = x^2$. Therefore both F and G are antiderivatives of f. Indeed, any function of the form $H(x) = \frac{1}{3}x^3 + C$, where C is a constant, is an antiderivative of f. The following theorem says that f has no other antiderivative. A proof of Theorem 1, using the Mean Value Theorem, is outlined in Exercise 46.

> **(1) Theorem** If F is an antiderivative of f on an interval I, then the most general antiderivative of f on I is
> $$F(x) + C$$
> where C is an arbitrary constant.

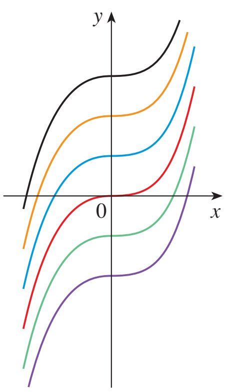


FIGURE 1
Members of the family of antiderivatives of $f(x) = x^2$

Going back to the function $f(x) = x^2$, we see that the general antiderivative of f is $\frac{1}{3}x^3 + C$. By assigning specific values to the constant C, we obtain a family of functions whose graphs are vertical translates of one another (see Figure 1). This makes sense because each curve must have the same slope at any given value of x.

EXAMPLE 1 | Find the most general antiderivative of each of the following functions.

(a) $f(x) = \sin x$ (b) $f(x) = 1/x$ (c) $f(x) = x^n$, $n \neq -1$

SOLUTION

(a) If $F(x) = -\cos x$, then $F'(x) = \sin x$, so an antiderivative of $\sin x$ is $-\cos x$. By Theorem 1, the most general antiderivative is $G(x) = -\cos x + C$.

(b) Recall from Section 3.7 that

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

So on the interval $(0, \infty)$ the general antiderivative of $1/x$ is $\ln x + C$. We also learned that

$$\frac{d}{dx}(\ln|x|) = \frac{1}{x}$$

for all $x \neq 0$. Theorem 1 then tells us that the general antiderivative of $f(x) = 1/x$ is $\ln|x| + C$ on any interval that doesn't contain 0. In particular, this is true on each of the intervals $(-\infty, 0)$ and $(0, \infty)$. So the general antiderivative of f is

$$F(x) = \begin{cases} \ln x + C_1 & \text{if } x > 0 \\ \ln(-x) + C_2 & \text{if } x < 0 \end{cases}$$

(c) We use the Power Rule to discover an antiderivative of x^n. In fact, if $n \neq -1$, then

$$\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = \frac{(n+1)x^n}{n+1} = x^n$$

Thus the general antiderivative of $f(x) = x^n$ is

$$F(x) = \frac{x^{n+1}}{n+1} + C$$

This is valid for $n \geqslant 0$ since then $f(x) = x^n$ is defined on an interval. If n is negative (but $n \neq -1$), it is valid on any interval that doesn't contain 0. ■

As in Example 1, every differentiation formula, when read from right to left, gives rise to an antidifferentiation formula. In Table 2 we list some particular antiderivatives. Each formula in the table is true because the derivative of the function in the right column appears in the left column. In particular, the first formula says that the antiderivative of a constant times a function is the constant times the antiderivative of the function. The second formula says that the antiderivative of a sum is the sum of the antiderivatives. (We use the notation $F' = f$, $G' = g$.)

To obtain the most general antiderivative from the particular ones in Table 2, we have to add a constant (or constants), as in Example 1.

(2) Table of Antidifferentiation Formulas

Function	Particular antiderivative	Function	Particular antiderivative
$cf(x)$	$cF(x)$	$\cos x$	$\sin x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sin x$	$-\cos x$
$x^n \; (n \neq -1)$	$\dfrac{x^{n+1}}{n+1}$	$\sec^2 x$	$\tan x$
$1/x$	$\ln\lvert x\rvert$	$\sec x \tan x$	$\sec x$
e^x	e^x	$\dfrac{1}{1+x^2}$	$\tan^{-1} x$
e^{cx}	$\dfrac{1}{c}e^{cx}$		

EXAMPLE 2 | Find all functions g such that

$$g'(x) = 4\sin x + \frac{2x^5 - \sqrt{x}}{x}$$

SOLUTION We first rewrite the given function as follows:

$$g'(x) = 4\sin x + \frac{2x^5}{x} - \frac{\sqrt{x}}{x} = 4\sin x + 2x^4 - \frac{1}{\sqrt{x}}$$

Thus we want to find an antiderivative of

$$g'(x) = 4\sin x + 2x^4 - x^{-1/2}$$

Using the formulas in Table 2 together with Theorem 1, we obtain

$$\begin{aligned} g(x) &= 4(-\cos x) + 2\frac{x^5}{5} - \frac{x^{1/2}}{\frac{1}{2}} + C \\ &= -4\cos x + \tfrac{2}{5}x^5 - 2\sqrt{x} + C \end{aligned}$$

In applications of calculus it is very common to have a situation as in Example 2, where it is required to find a function, given knowledge about its derivatives. An equation that involves the derivatives of a function is called a **differential equation**. These will be studied in some detail in Chapter 7, but for the present we can solve some elementary differential equations. The general solution of a differential equation involves an arbitrary constant (or constants) as in Example 2. However, there may be some extra conditions given that will determine the constants and therefore uniquely specify the solution.

A differential equation of the form

$$\frac{dy}{dt} = f(t)$$

is called a **pure-time differential equation** because the right side of the equation does not depend on y; it depends only on t (time). The solution will be a family of antiderivatives of f. The initial value of the solution may be specified by an **initial condition** of the form $y = y_0$ when $t = t_0$. Then the problem of finding a solution of the differential equation that also satisfies the initial condition is called an **initial-value problem**:

$$\frac{dy}{dt} = f(t) \qquad y = y_0 \text{ when } t = t_0$$

Figure 2 shows the graphs of the function f' in Example 3 and its antiderivative f. Notice that $f'(x) > 0$, so f is always increasing. Also notice that when f' has a maximum or minimum, f appears to have an inflection point. So the graph serves as a check on our calculation.

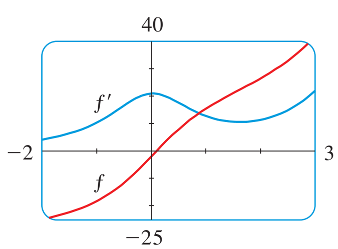


FIGURE 2

EXAMPLE 3 | Find f if $f'(x) = e^x + 20(1 + x^2)^{-1}$ and $f(0) = -2$.

SOLUTION The general antiderivative of

$$f'(x) = e^x + \frac{20}{1 + x^2}$$

is

$$f(x) = e^x + 20 \tan^{-1}x + C$$

To determine C we use the fact that $f(0) = -2$:

$$f(0) = e^0 + 20 \tan^{-1} 0 + C = -2$$

Thus we have $C = -2 - 1 = -3$, so the particular solution is

$$f(x) = e^x + 20 \tan^{-1}x - 3$$

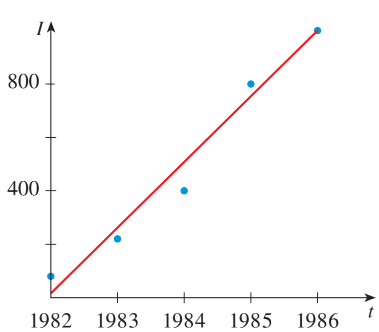


FIGURE 3

EXAMPLE 4 | **HIV incidence and prevalence** The rate I at which people were becoming infected with HIV (termed the incidence) in New York in the early 1980s is plotted in Figure 3. We can see from the figure that the data are well approximated by the linear function $I(t) = 16 + 242t$, where t measures the number of years since 1982. Suppose there were 80 infections at year $t = 0$. What is the number of infections expected in 1990 (termed the prevalence)?

SOLUTION Let $P(t)$ be the prevalence in year t, that is, the number of infections. We are given that

$$\frac{dP}{dt} = I(t) = 16 + 242t \qquad P(0) = 80$$

This is an initial-value problem for a pure-time differential equation. The general solution is given by the antiderivative of dP/dt:

$$P(t) = 16t + 121t^2 + C$$

Then $P(0) = C$, but we are given that $P(0) = 80$, so $C = 80$. The solution is

$$P(t) = 80 + 16t + 121t^2$$

The projected number of infections in 1990 is

$$P(8) = 80 + 16 \cdot 8 + 121 \cdot 8^2 = 7952$$

The actual number was estimated to be 7200. ■

EXAMPLE 5 | Find f if $f''(x) = 12x^2 + 6x - 4$, $f(0) = 4$, and $f(1) = 1$.

SOLUTION The general antiderivative of $f''(x) = 12x^2 + 6x - 4$ is

$$f'(x) = 12\frac{x^3}{3} + 6\frac{x^2}{2} - 4x + C = 4x^3 + 3x^2 - 4x + C$$

Using the antidifferentiation rules once more, we find that

$$f(x) = 4\frac{x^4}{4} + 3\frac{x^3}{3} - 4\frac{x^2}{2} + Cx + D = x^4 + x^3 - 2x^2 + Cx + D$$

To determine C and D we use the given conditions that $f(0) = 4$ and $f(1) = 1$. Since $f(0) = 0 + D = 4$, we have $D = 4$. Since

$$f(1) = 1 + 1 - 2 + C + 4 = 1$$

we have $C = -3$. Therefore the required function is

$$f(x) = x^4 + x^3 - 2x^2 - 3x + 4$$ ■

Antidifferentiation is particularly useful in analyzing the motion of an object moving in a straight line. Recall that if the object has position function $s = f(t)$, then the velocity function is $v(t) = s'(t)$. This means that the position function is an antiderivative of the velocity function. Likewise, the acceleration function is $a(t) = v'(t)$, so the velocity function is an antiderivative of the acceleration. If the acceleration and the initial values $s(0)$ and $v(0)$ are known, then the position function can be found by antidifferentiating twice.

EXAMPLE 6 | A particle moves in a straight line and has acceleration given by $a(t) = 6t + 4$. Its initial velocity is $v(0) = -6$ cm/s and its initial displacement is $s(0) = 9$ cm. Find its position function $s(t)$.

SOLUTION Since $v'(t) = a(t) = 6t + 4$, antidifferentiation gives

$$v(t) = 6\frac{t^2}{2} + 4t + C = 3t^2 + 4t + C$$

Note that $v(0) = C$. But we are given that $v(0) = -6$, so $C = -6$ and

$$v(t) = 3t^2 + 4t - 6$$

Since $v(t) = s'(t)$, s is the antiderivative of v:

$$s(t) = 3\frac{t^3}{3} + 4\frac{t^2}{2} - 6t + D = t^3 + 2t^2 - 6t + D$$

This gives $s(0) = D$. We are given that $s(0) = 9$, so $D = 9$ and the required position function is

$$s(t) = t^3 + 2t^2 - 6t + 9$$ ■

EXERCISES 4.6

1–20 Find the most general antiderivative of the function. (Check your answer by differentiation.)

1. $f(x) = \frac{1}{2} + \frac{3}{4}x^2 - \frac{4}{5}x^3$

2. $f(x) = 8x^9 - 3x^6 + 12x^3$

3. $f(x) = (x + 1)(2x - 1)$

4. $f(x) = x(2 - x)^2$

5. $f(x) = 5x^{1/4} - 7x^{3/4}$

6. $f(x) = 2x + 3x^{1.7}$

7. $f(x) = 6\sqrt{x} - \sqrt[6]{x}$

8. $f(x) = \sqrt[4]{x^3} + \sqrt[3]{x^4}$

9. $f(x) = \sqrt{2}$

10. $f(x) = e^2$

11. $c(t) = \dfrac{3}{t^2},\ t > 0$

12. $h(m) = \dfrac{2}{\sqrt{m}}$

13. $g(\theta) = \cos\theta - 5\sin\theta$

14. $f(x) = 2\sqrt{x} + 6\cos x$

15. $v(s) = 4s + 3e^s$

16. $u(r) = e^{-2r}$

17. $f(u) = \dfrac{u^4 + 3\sqrt{u}}{u^2}$

18. $f(x) = 3e^x + 7\sec^2 x$

19. $f(t) = \dfrac{t^4 - t^2 + 1}{t^2}$

20. $f(x) = \dfrac{1 + x - x^2}{x}$

21–28 Solve the initial-value problem.

21. $\dfrac{dy}{dt} = t^2 + 1,\quad t \geqslant 0,\quad y = 6$ when $t = 0$

22. $\dfrac{dy}{dt} = 1 + \dfrac{2}{t},\quad t > 0,\quad y = 5$ when $t = 1$

23. $\dfrac{dP}{dt} = 2e^{3t},\quad t \geqslant 0,\quad P(0) = 1$

24. $\dfrac{dm}{dt} = 100e^{-0.4t},\quad t \geqslant 0,\quad m(0) = 50$

25. $\dfrac{dr}{d\theta} = \cos\theta + \sec\theta\tan\theta,\quad 0 < \theta < \pi/2,\quad r(\pi/3) = 4$

26. $\dfrac{dy}{dx} = x^2 + 1 + \dfrac{1}{x^2 + 1},\quad y = 0$ when $x = 1$

27. $\dfrac{du}{dt} = \sqrt{t} + \dfrac{2}{\sqrt{t}},\quad t > 0,\quad u(1) = 5$

28. $\dfrac{dv}{dt} = e^{-t}(1 + e^{2t}),\quad t \geqslant 0,\quad v(0) = 3$

29–40 Find f.

29. $f''(x) = 6x + 12x^2$

30. $f''(x) = 2 + x^3 + x^6$

31. $f''(x) = \frac{2}{3}x^{2/3}$

32. $f''(x) = 6x + \sin x$

33. $f'(x) = 1 - 6x,\quad f(0) = 8$

34. $f'(x) = 8x^3 + 12x + 3,\quad f(1) = 6$

35. $f'(x) = \sqrt{x}(6 + 5x),\quad f(1) = 10$

36. $f'(x) = 2x - 3/x^4,\quad x > 0,\quad f(1) = 3$

37. $f''(\theta) = \sin\theta + \cos\theta,\quad f(0) = 3,\quad f'(0) = 4$

38. $f''(x) = 8x^3 + 5,\quad f(1) = 0,\quad f'(1) = 8$

39. $f''(x) = 2 - 12x,\quad f(0) = 9,\quad f(2) = 15$

40. $f''(t) = 2e^t + 3\sin t,\quad f(0) = 0,\quad f(\pi) = 0$

41. Bacteria culture A culture of the bacterium *Rhodobacter sphaeroides* initially has 25 bacteria and t hours later increases at a rate of $3.4657e^{0.1386t}$ bacteria per hour. Find the population size after four hours.

42. A sample of cesium-37 with an initial mass of 75 mg decays t years later at a rate of $1.7325e^{-0.0231t}$ mg/year. Find the mass of the sample after 20 years.

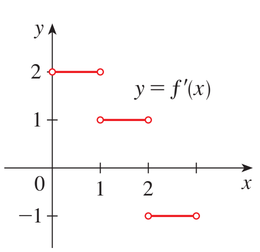
43. A particle moves along a straight line with velocity function $v(t) = \sin t - \cos t$ and its initial displacement is $s(0) = 0$ m. Find its position function $s(t)$.

44. A particle moves with acceleration function $a(t) = 5 + 4t - 2t^2$. Its initial velocity is $v(0) = 3$ m/s and its initial displacement is $s(0) = 10$ m. Find its position after t seconds.

45. A stone is dropped from the upper observation deck (the Space Deck) of the CN Tower, 450 m above the ground.
(a) Find the distance of the stone above ground level at time t. Use the fact that acceleration due to gravity is $g \approx 9.8$ m/s^2.
(b) How long does it take the stone to reach the ground?
(c) With what velocity does it strike the ground?

46. To prove Theorem 1, let F and G be any two antiderivatives of f on I and let $H = G - F$.
(a) If x_1 and x_2 are any two numbers in I with $x_1 < x_2$, apply the Mean Value Theorem on the interval $[x_1, x_2]$ to show that $H(x_1) = H(x_2)$. Why does this show that H is a constant function?
(b) Deduce Theorem 1 from the result of part (a).

47. The graph of f' is shown in the figure. Sketch the graph of f if f is continuous on $[0, 3]$ and $f(0) = -1$.



48. Find a function f such that $f'(x) = x^3$ and the line $x + y = 0$ is tangent to the graph of f.

Chapter 4 REVIEW

CONCEPT CHECK

1. Explain the difference between an absolute maximum and a local maximum. Illustrate with a sketch.
2. (a) What does the Extreme Value Theorem say?
   (b) Explain how the Closed Interval Method works.
3. (a) State Fermat's Theorem.
   (b) Define a critical number of f.
4. State the Mean Value Theorem and give a geometric interpretation.
5. (a) State the Increasing/Decreasing Test.
   (b) What does it mean to say that f is concave upward on an interval I?
   (c) State the Concavity Test.
   (d) What are inflection points? How do you find them?
6. (a) State the First Derivative Test.
   (b) State the Second Derivative Test.
   (c) What are the relative advantages and disadvantages of these tests?
7. (a) What does l'Hospital's Rule say?
   (b) How can you use l'Hospital's Rule if you have a product $f(x)g(x)$ where $f(x) \to 0$ and $g(x) \to \infty$ as $x \to a$?
   (c) How can you use l'Hospital's Rule if you have a difference $f(x) - g(x)$ where $f(x) \to \infty$ and $g(x) \to \infty$ as $x \to a$?
8. (a) What is an equilibrium of the recursive sequence $x_{t+1} = f(x_t)$?
   (b) What is a stable equilibrium? An unstable equilibrium?
   (c) State the Stability Criterion.
9. (a) What is an antiderivative of a function f?
   (b) Suppose F_1 and F_2 are both antiderivatives of f on an interval I. How are F_1 and F_2 related?

Answers to the Concept Check can be found on the back endpapers.

TRUE-FALSE QUIZ

Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give an example that disproves the statement.

1. If $f'(c) = 0$, then f has a local maximum or minimum at c.
2. If f has an absolute minimum value at c, then $f'(c) = 0$.
3. If f is continuous on (a, b), then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in (a, b).
4. If f is differentiable and $f(-1) = f(1)$, then there is a number c such that $|c| < 1$ and $f'(c) = 0$.
5. If $f'(x) < 0$ for $1 < x < 6$, then f is decreasing on $(1, 6)$.
6. If $f''(2) = 0$, then $(2, f(2))$ is an inflection point of the curve $y = f(x)$.
7. If $f'(x) = g'(x)$ for $0 < x < 1$, then $f(x) = g(x)$ for $0 < x < 1$.
8. There exists a function f such that $f(1) = -2$, $f(3) = 0$, and $f'(x) > 1$ for all x.
9. There exists a function f such that $f(x) > 0$, $f'(x) < 0$, and $f''(x) > 0$ for all x.
10. There exists a function f such that $f(x) < 0$, $f'(x) < 0$, and $f''(x) > 0$ for all x.
11. If f and g are increasing on an interval I, then $f + g$ is increasing on I.
12. If f and g are increasing on an interval I, then $f - g$ is increasing on I.
13. If f and g are increasing on an interval I, then fg is increasing on I.
14. If f and g are positive increasing functions on an interval I, then fg is increasing on I.
15. If f is increasing and $f(x) > 0$ on I, then $g(x) = 1/f(x)$ is decreasing on I.
16. If f is even, then f' is even.
17. If f is periodic, then f' is periodic.
18. $\lim_{x \to 0} \dfrac{x}{e^x} = 1$

EXERCISES

1–6 Find the local and absolute extreme values of the function on the given interval.

1. $f(x) = x^3 - 6x^2 + 9x + 1, \quad [2, 4]$

2. $f(x) = x\sqrt{1 - x}, \quad [-1, 1]$

3. $f(x) = \dfrac{3x - 4}{x^2 + 1}, \quad [-2, 2]$

4. $f(x) = (x^2 + 2x)^3, \quad [-2, 1]$

5. $f(x) = x + \sin 2x, \quad [0, \pi]$

6. $f(x) = (\ln x)/x^2, \quad [1, 3]$

7–14

(a) Find the vertical and horizontal asymptotes, if any.
(b) Find the intervals of increase or decrease.
(c) Find the local maximum and minimum values.
(d) Find the intervals of concavity and the inflection points.
(e) Use the information from parts (a)–(d) to sketch the graph of f. Check your work with a graphing device.

7. $f(x) = 2 - 2x - x^3$

8. $f(x) = x^4 + 4x^3$

9. $f(x) = x + \sqrt{1 - x}$

10. $f(x) = \dfrac{1}{1 - x^2}$

11. $y = \sin^2 x - 2\cos x$

12. $y = e^{2x - x^2}$

13. $y = e^x + e^{-3x}$

14. $y = \ln(x^2 - 1)$

15. Antibiotic pharmacokinetics A model for the concentration of an antibiotic drug in the bloodstream t hours after the administration of the drug is

$$C(t) = 2.5(e^{-0.3t} - e^{-0.7t})$$

where C is measured in μg/mL.
(a) At what time does the concentration have its maximum value? What is the maximum value?
(b) At what time does the inflection point occur? What is the significance of the inflection point?

16. Drug pharmacokinetics Another model for the concentration of a drug in the bloodstream is

$$C(t) = 0.5t^2 e^{-0.6t}$$

where t is measured in hours and C is measured in μg/mL.
(a) At what time does the concentration have its largest value? What is the largest value?
(b) How many inflection points are there? At what times do they occur? What is the significance of each inflection point?
(c) Compare the graphs of this concentration function and the one in Exercise 15. How are the graphs similar? How are they different?

17. Population bound Suppose that an initial population size is 300 individuals and the population grows at a rate of at most 120 individuals per week. What can you say about the population size after five weeks?

18. Sketch the graph of a function that satisfies the following conditions:

$f(0) = 0$, $\quad f$ is continuous and even,
$f'(x) = 2x$ if $0 < x < 1$, $\quad f'(x) = -1$ if $1 < x < 3$,
$f'(x) = 1$ if $x > 3$

19–25 Evaluate the limit.

19. $\displaystyle\lim_{x \to 0} \frac{\tan \pi x}{\ln(1 + x)}$

20. $\displaystyle\lim_{x \to 0} \frac{1 - \cos x}{x^2 + x}$

21. $\displaystyle\lim_{x \to 0} \frac{e^{4x} - 1 - 4x}{x^2}$

22. $\displaystyle\lim_{x \to \infty} \frac{e^{4x} - 1 - 4x}{x^2}$

23. $\displaystyle\lim_{x \to \infty} x^3 e^{-x}$

24. $\displaystyle\lim_{x \to 0^+} x^2 \ln x$

25. $\displaystyle\lim_{x \to 1^+} \left(\frac{x}{x - 1} - \frac{1}{\ln x} \right)$

26. Rank the functions in order of how quickly they grow as $x \to \infty$.

$y = \sqrt[4]{x} \qquad y = \ln(10x) \qquad y = 10^x \qquad y = \sqrt{1 + e^x}$

27. Find two positive integers such that the sum of the first number and four times the second number is 1000 and the product of the numbers is as large as possible.

28. Find the point on the hyperbola $xy = 8$ that is closest to the point $(3, 0)$.

29. The velocity of a wave of length L in deep water is

$$v = K\sqrt{\frac{L}{C} + \frac{C}{L}}$$

where K and C are known positive constants. What is the length of the wave that gives the minimum velocity?

30. The **Ricker model** for population growth is a discrete-time model of the form

$$n_{t+1} = cn_t e^{-\lambda n_t}$$

For the constants $c = 2$ and $\lambda = 3$, the model is $n_{t+1} = f(n_t)$, where the updating function is

$$f(n) = 2ne^{-3n}$$

Find the largest value of f and interpret it.

31. Drug resistance evolution A simple model for the spread of drug resistance is given by $\Delta p = p(1 - p)s$, where s is a measure of the reproductive advantage of the drug resistance gene in the presence of drugs, p is the frequency of the drug resistance gene in the population, and Δp is the change in the frequency of the drug resistance gene in the population after one year. Notice that the amount of change in the frequency Δp differs depending on the gene's current frequency p. What current frequency makes the rate of evolution Δp the largest?

32. The **thermic effect of food** (TEF) is the increase in metabolic rate after a meal. Researchers used the functions

$$f(t) = 175.9te^{-t/1.3} \qquad g(t) = 113.6te^{-t/1.85}$$

to model the TEF (measured in kJ/h) for a lean person and an obese person, respectively.

(a) Find the maximum value of the TEF for both individuals.

(b) Graph the TEF functions for both individuals. Describe how the graphs are similar and how they differ.

Source: Adapted from G. Reed et al., "Measuring the Thermic Effects of Food," *American Journal of Clinical Nutrition* 63 (1996): 164–69.

33–34 Find the equilibria of the difference equation and classify them as stable or unstable.

33. $x_{t+1} = \dfrac{4x_t}{1 + 5x_t}$

34. $x_{t+1} = 5x_t e^{-4x_t}$

35. Find the equilibria of the difference equation

$$x_{t+1} = \frac{6x_t^2}{x_t^2 + 8}$$

and classify them as stable or unstable. Use cobwebbing to evaluate $\lim_{t\to\infty} x_t$ for $x_0 = 1$ and $x_0 = 3$.

36. Let $f(x) = 1.07x + \sin x$, $0 \leqslant x \leqslant 11$. How many equilibria does the recursion $x_{t+1} = f(x_t)$ have? Estimate their values and explain why they are stable or unstable.

37–40 Find the most general antiderivative of the function.

37. $f(x) = \sin x + \sec x \tan x, \quad 0 \leqslant x < \pi/2$

38. $g(t) = (1 + t)/\sqrt{t}$

39. $q(t) = 2 + (t + 1)(t^2 - 1)$

40. $w(\theta) = 2\theta - 3\cos\theta$

41–42 Solve the initial-value problem.

41. $\dfrac{dy}{dt} = 1 - e^{\pi t}, \quad y = 0$ when $t = 0$

42. $\dfrac{dr}{dt} = \dfrac{4}{1 + t^2}, \quad r(1) = 2$

43–44 Find $f(x)$.

43. $f''(x) = 1 - 6x + 48x^2, \quad f(0) = 1, \quad f'(0) = 2$

44. $f''(x) = 2x^3 + 3x^2 - 4x + 5, \quad f(0) = 2, \quad f(1) = 0$

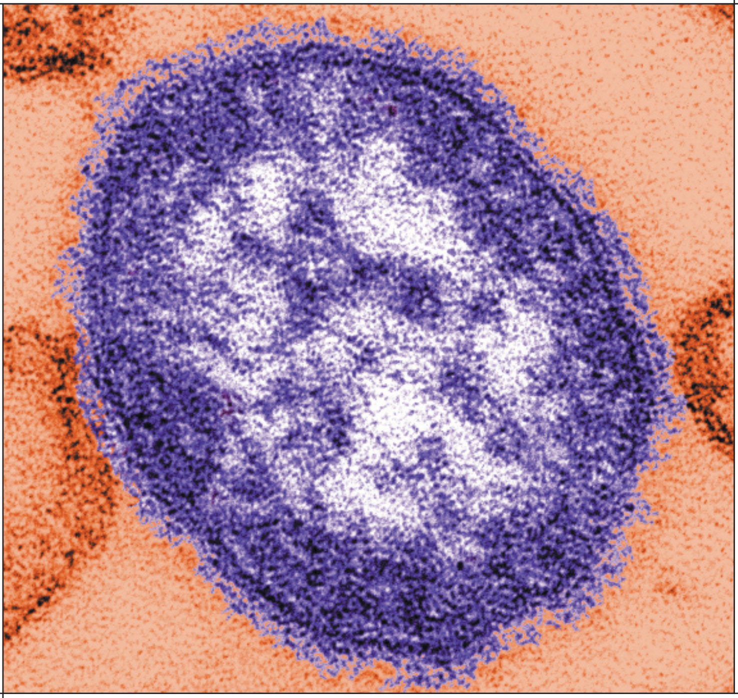
45. A particle moves in a straight line with acceleration $a(t) = \sin t + 3\cos t$, initial displacement $s(0) = 0$, and initial velocity $v(0) = 2$. Find its position function $s(t)$.

46. Sketch the graph of a continuous, even function f such that $f(0) = 0$, $f'(x) = 2x$ if $0 < x < 1$, $f'(x) = -1$ if $1 < x < 3$, and $f'(x) = 1$ if $x > 3$.

47. If a rectangle has its base on the x-axis and two vertices on the curve $y = e^{-x^2}$, show that the rectangle has the largest possible area when the two vertices are at the points of inflection of the curve.

Integrals

5



Shown is a transmission electron micrograph of a measles virus. In Sections 5.1 and 5.3 we will see how an integral can be used to represent the total amount of infection needed to develop symptoms of measles.

Scott Camazine / Alamy

IN CHAPTER 3 WE USED the problems of finding tangent lines and rates of increase to introduce the derivative, which is the central idea in differential calculus. In much the same way, this chapter starts with the area and distance problems and uses them to formulate the idea of a definite integral, which is the basic concept of integral calculus. We will see in this chapter and the next how to use integrals to solve problems concerning disease development, population dynamics, biological control, blood flow, and cardiac output, among others.

There is a connection between integral calculus and differential calculus. The Fundamental Theorem of Calculus relates the integral to the derivative, and we will see in this chapter that it greatly simplifies the solution of many problems.

5.1 Areas, Distances, and Pathogenesis

In this section we discover that in trying to find the area under a curve or the distance traveled by a car, we end up with the same special type of limit.

Why would a biologist be interested in calculating an area? A botanist might want to know the area of a leaf and compare it with the leaf's area at other stages of its development. An ecologist might want to know the area of a lake and compare it with the area in previous years. An oncologist might want to know the area of a tumor and compare it with the areas at prior times to see how quickly it is growing. But there are also indirect ways in which areas are of interest. We will see at the end of this section, for example, that the area beneath part of the pathogenesis curve for a measles infection represents the amount of infection needed to develop symptoms of the disease.

The Area Problem

We begin by attempting to solve the *area problem*: Find the area of the region S that lies under the curve $y = f(x)$ from a to b. This means that S, illustrated in Figure 1, is bounded by the graph of a continuous function f [where $f(x) \geqslant 0$], the vertical lines $x = a$ and $x = b$, and the x-axis.

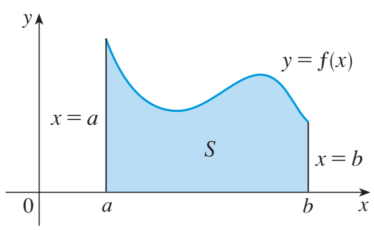


FIGURE 1
$S = \{(x,y) \mid a \leqslant x \leqslant b, 0 \leqslant y \leqslant f(x)\}$

In trying to solve the area problem we have to ask ourselves: What is the meaning of the word *area*? This question is easy to answer for regions with straight sides. For a rectangle, the area is defined as the product of the length and the width. The area of a triangle is half the base times the height. The area of a polygon is found by dividing it into triangles (as in Figure 2) and adding the areas of the triangles.

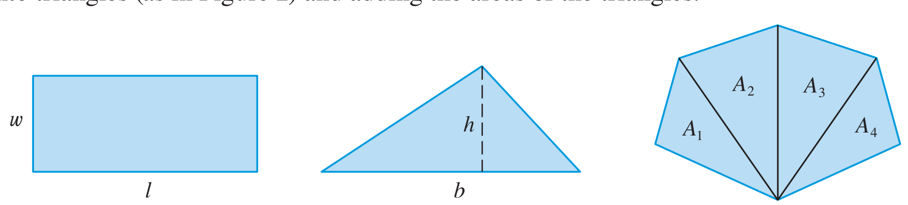


FIGURE 2 $A = lw$ $\quad A = \frac{1}{2}bh \quad A = A_1 + A_2 + A_3 + A_4$

However, it isn't so easy to find the area of a region with curved sides. We all have an intuitive idea of what the area of a region is. But part of the area problem is to make this intuitive idea precise by giving an exact definition of area.

Recall that in defining a tangent we first approximated the slope of the tangent line by slopes of secant lines and then we took the limit of these approximations. We pursue a similar idea for areas. We first approximate the region S by rectangles and then we take the limit of the areas of these rectangles as we increase the number of rectangles. The following example illustrates the procedure.

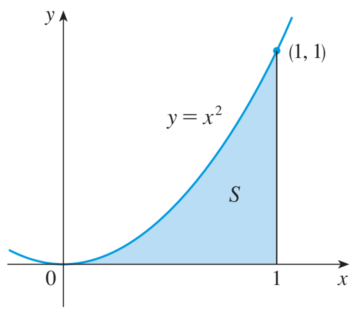


FIGURE 3

EXAMPLE 1 | Use rectangles to estimate the area under the parabola $y = x^2$ from 0 to 1 (the parabolic region S illustrated in Figure 3).

SOLUTION We first notice that the area of S must be somewhere between 0 and 1 because S is contained in a square with side length 1, but we can certainly do better than that. Suppose we divide S into four strips S_1, S_2, S_3, and S_4 by drawing the vertical lines $x = \frac{1}{4}$, $x = \frac{1}{2}$, and $x = \frac{3}{4}$ as in Figure 4(a).

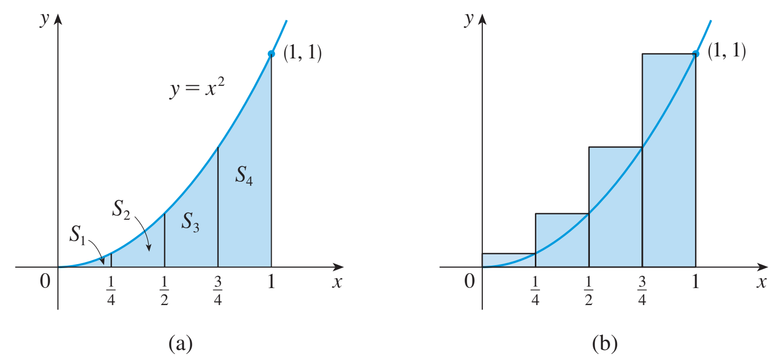


FIGURE 4

We can approximate each strip by a rectangle that has the same base as the strip and whose height is the same as the right edge of the strip [see Figure 4(b)]. In other words, the heights of these rectangles are the values of the function $f(x) = x^2$ at the right endpoints of the subintervals $\left[0, \frac{1}{4}\right]$, $\left[\frac{1}{4}, \frac{1}{2}\right]$, $\left[\frac{1}{2}, \frac{3}{4}\right]$, and $\left[\frac{3}{4}, 1\right]$.

Each rectangle has width $\frac{1}{4}$ and the heights are $\left(\frac{1}{4}\right)^2$, $\left(\frac{1}{2}\right)^2$, $\left(\frac{3}{4}\right)^2$, and 1^2. If we let R_4 be the sum of the areas of these approximating rectangles, we get

$$R_4 = \tfrac{1}{4} \cdot \left(\tfrac{1}{4}\right)^2 + \tfrac{1}{4} \cdot \left(\tfrac{1}{2}\right)^2 + \tfrac{1}{4} \cdot \left(\tfrac{3}{4}\right)^2 + \tfrac{1}{4} \cdot 1^2 = \tfrac{15}{32} = 0.46875$$

From Figure 4(b) we see that the area A of S is less than R_4, so

$$A < 0.46875$$

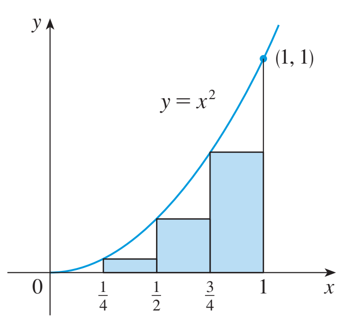


FIGURE 5

Instead of using the rectangles in Figure 4(b) we could use the smaller rectangles in Figure 5 whose heights are the values of f at the *left* endpoints of the subintervals. (The leftmost rectangle has collapsed because its height is 0.) The sum of the areas of these approximating rectangles is

$$L_4 = \tfrac{1}{4} \cdot 0^2 + \tfrac{1}{4} \cdot \left(\tfrac{1}{4}\right)^2 + \tfrac{1}{4} \cdot \left(\tfrac{1}{2}\right)^2 + \tfrac{1}{4} \cdot \left(\tfrac{3}{4}\right)^2 = \tfrac{7}{32} = 0.21875$$

We see that the area of S is larger than L_4, so we have lower and upper estimates for A:

$$0.21875 < A < 0.46875$$

We can repeat this procedure with a larger number of strips. Figure 6 shows what happens when we divide the region S into eight strips of equal width.

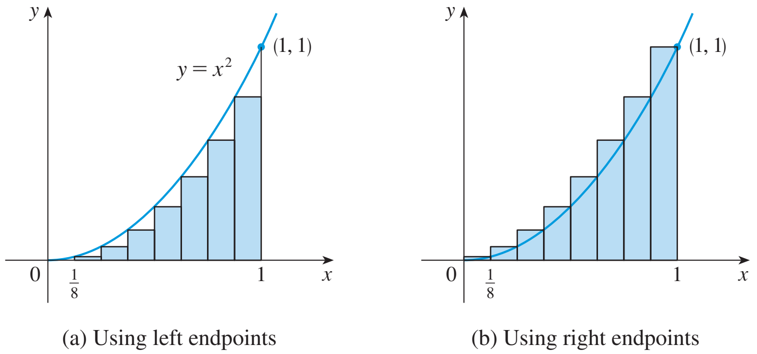


FIGURE 6
Approximating S with eight rectangles

By computing the sum of the areas of the smaller rectangles (L_8) and the sum of the areas of the larger rectangles (R_8), we obtain better lower and upper estimates for A:

$$0.2734375 < A < 0.3984375$$

n	L_n	R_n
10	0.2850000	0.3850000
20	0.3087500	0.3587500
30	0.3168519	0.3501852
50	0.3234000	0.3434000
100	0.3283500	0.3383500
1000	0.3328335	0.3338335

So one possible answer to the question is to say that the true area of S lies somewhere between 0.2734375 and 0.3984375.

We could obtain better estimates by increasing the number of strips. The table at the left shows the results of similar calculations (with a computer) using n rectangles whose heights are found with left endpoints (L_n) or right endpoints (R_n). In particular, we see by using 50 strips that the area lies between 0.3234 and 0.3434. With 1000 strips we narrow it down even more: A lies between 0.3328335 and 0.3338335. A good estimate is obtained by averaging these numbers: $A \approx 0.3333335$. ■

From the values in the table in Example 1, it looks as if R_n is approaching $\frac{1}{3}$ as n increases. We confirm this in the next example.

EXAMPLE 2 | For the region S in Example 1, show that the sum of the areas of the upper approximating rectangles approaches $\frac{1}{3}$, that is,

$$\lim_{n\to\infty} R_n = \tfrac{1}{3}$$

SOLUTION R_n is the sum of the areas of the n rectangles in Figure 7. Each rectangle has width $1/n$ and the heights are the values of the function $f(x) = x^2$ at the points $1/n, 2/n, 3/n, \ldots, n/n$; that is, the heights are $(1/n)^2, (2/n)^2, (3/n)^2, \ldots, (n/n)^2$. Thus

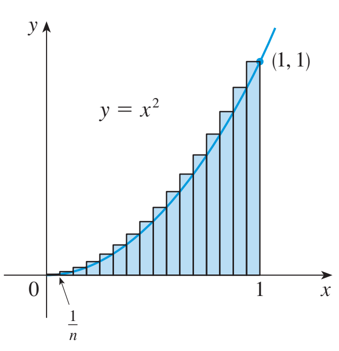


FIGURE 7

$$\begin{aligned} R_n &= \frac{1}{n}\left(\frac{1}{n}\right)^2 + \frac{1}{n}\left(\frac{2}{n}\right)^2 + \frac{1}{n}\left(\frac{3}{n}\right)^2 + \cdots + \frac{1}{n}\left(\frac{n}{n}\right)^2 \\ &= \frac{1}{n}\cdot\frac{1}{n^2}(1^2 + 2^2 + 3^2 + \cdots + n^2) \\ &= \frac{1}{n^3}(1^2 + 2^2 + 3^2 + \cdots + n^2) \end{aligned}$$

Here we need the formula for the sum of the squares of the first n positive integers:

(1) $$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Perhaps you have seen this formula before. It is proved in Example 5 in Appendix F.

Putting Formula 1 into our expression for R_n, we get

$$R_n = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2}$$

Limits of sequences were introduced in Section 2.1.

Now we compute the limit of the sequence R_n:

$$\begin{aligned}\lim_{n\to\infty} R_n &= \lim_{n\to\infty} \frac{(n+1)(2n+1)}{6n^2} \\ &= \lim_{n\to\infty} \frac{1}{6}\left(\frac{n+1}{n}\right)\left(\frac{2n+1}{n}\right) \\ &= \lim_{n\to\infty} \frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right) \\ &= \frac{1}{6} \cdot 1 \cdot 2 = \frac{1}{3}\end{aligned}$$

It can be shown that the lower approximating sums also approach $\frac{1}{3}$, that is,

$$\lim_{n\to\infty} L_n = \tfrac{1}{3}$$

TEC In Visual 5.1 you can create pictures like those in Figures 8 and 9 for other values of n.

From Figures 8 and 9 it appears that, as n increases, both L_n and R_n become better and better approximations to the area of S. Therefore we *define* the area A to be the limit of the sums of the areas of the approximating rectangles, that is,

$$A = \lim_{n\to\infty} R_n = \lim_{n\to\infty} L_n = \tfrac{1}{3}$$

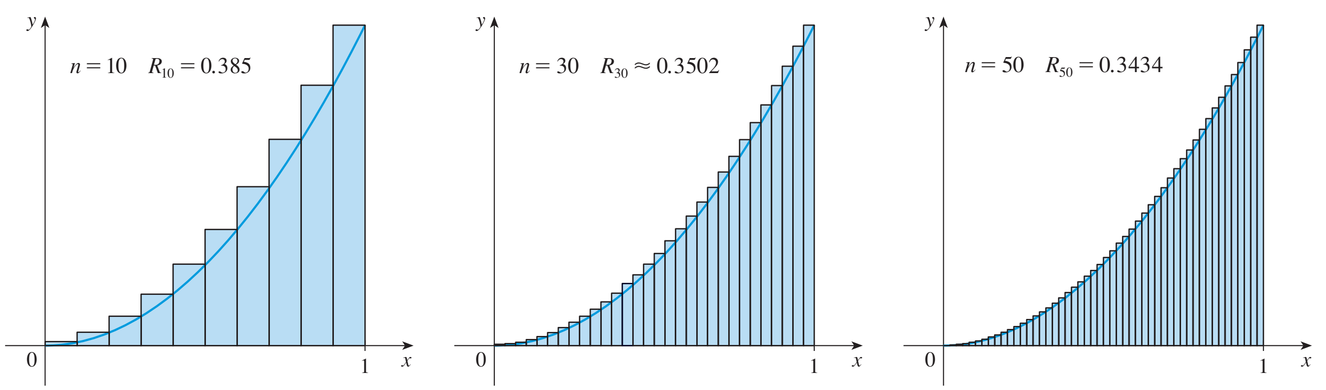


FIGURE 8

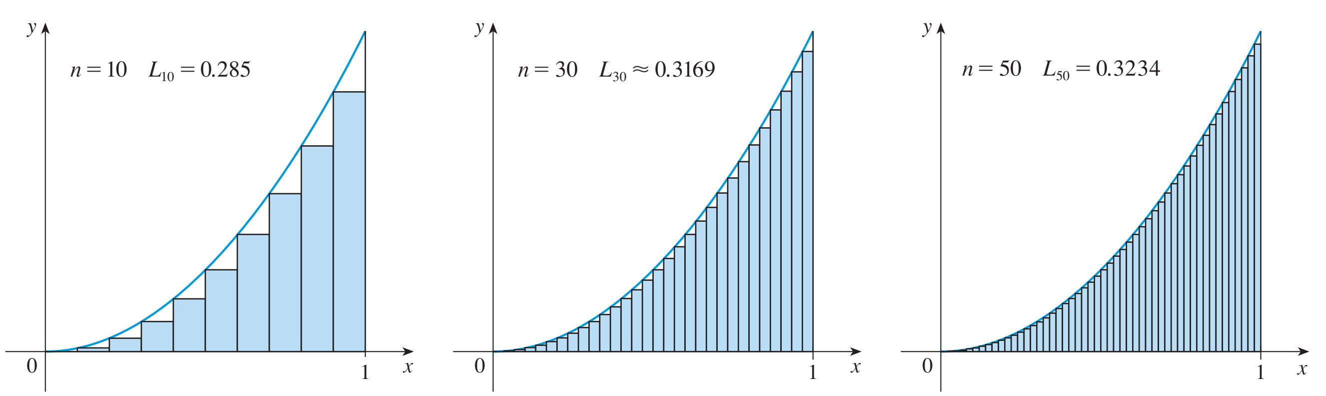


FIGURE 9 The area is the number that is smaller than all upper sums and larger than all lower sums.

Let's apply the idea of Examples 1 and 2 to the more general region S of Figure 1. We start by subdividing S into n strips $S_1, S_2, \ldots, S_n$ of equal width as in Figure 10.

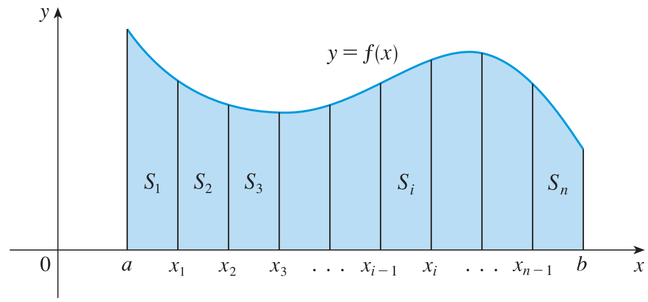


FIGURE 10

The width of the interval $[a, b]$ is $b - a$, so the width of each of the n strips is

$$\Delta x = \frac{b - a}{n}$$

These strips divide the interval $[a, b]$ into n subintervals

$$[x_0, x_1], \quad [x_1, x_2], \quad [x_2, x_3], \quad \ldots, \quad [x_{n-1}, x_n]$$

where $x_0 = a$ and $x_n = b$. The right endpoints of the subintervals are

$$\begin{aligned} x_1 &= a + \Delta x, \\ x_2 &= a + 2\,\Delta x, \\ x_3 &= a + 3\,\Delta x, \\ &\vdots \end{aligned}$$

Let's approximate the ith strip S_i by a rectangle with width Δx and height $f(x_i)$, which is the value of f at the right endpoint (see Figure 11). Then the area of the ith rectangle is $f(x_i)\,\Delta x$. What we think of intuitively as the area of S is approximated by the sum of the areas of these rectangles, which is

$$R_n = f(x_1)\,\Delta x + f(x_2)\,\Delta x + \cdots + f(x_n)\,\Delta x$$

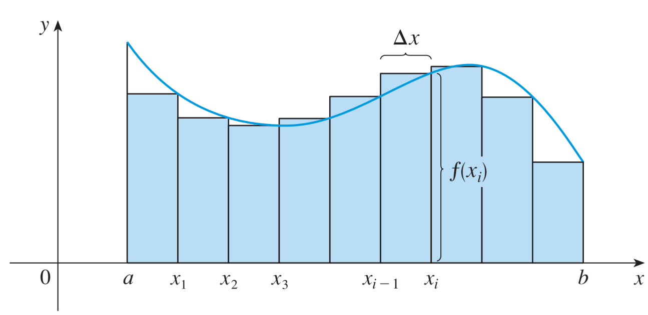
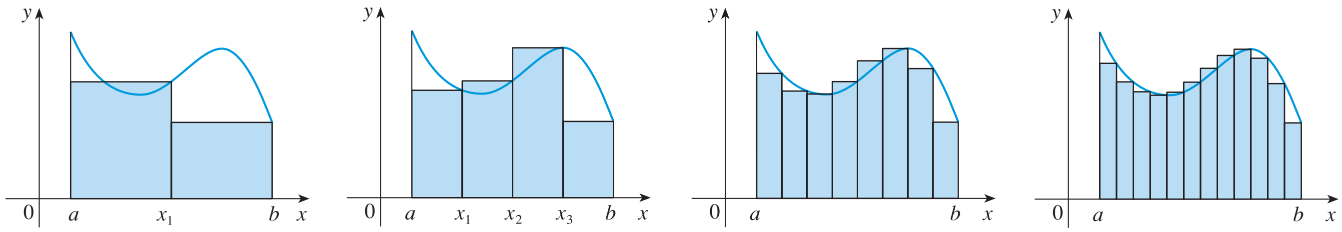


FIGURE 11

Figure 12 shows this approximation for $n = 2, 4, 8$, and 12. Notice that this approximation appears to become better and better as the number of strips increases, that is, as $n \to \infty$. Therefore we define the area A of the region S in the following way.



(a) $n = 2$ (b) $n = 4$ (c) $n = 8$ (d) $n = 12$

FIGURE 12

> **(2) Definition** The **area** A of the region S that lies under the graph of the continuous function f is the limit of the sum of the areas of approximating rectangles:
>
> $$A = \lim_{n \to \infty} R_n = \lim_{n \to \infty} [f(x_1)\,\Delta x + f(x_2)\,\Delta x + \cdots + f(x_n)\,\Delta x]$$

It can be proved that the limit in Definition 2 always exists, since we are assuming that f is continuous. It can also be shown that we get the same value if we use left endpoints:

(3) $$A = \lim_{n \to \infty} L_n = \lim_{n \to \infty} [f(x_0)\,\Delta x + f(x_1)\,\Delta x + \cdots + f(x_{n-1})\,\Delta x]$$

In fact, instead of using left endpoints or right endpoints, we could take the height of the ith rectangle to be the value of f at any number x_i^* in the ith subinterval $[x_{i-1}, x_i]$. We call the numbers $x_1^*, x_2^*, \ldots, x_n^*$ the **sample points**. Figure 13 shows approximating rectangles when the sample points are not chosen to be endpoints. So a more general expression for the area of S is

(4) $$A = \lim_{n \to \infty} [f(x_1^*)\,\Delta x + f(x_2^*)\,\Delta x + \cdots + f(x_n^*)\,\Delta x]$$

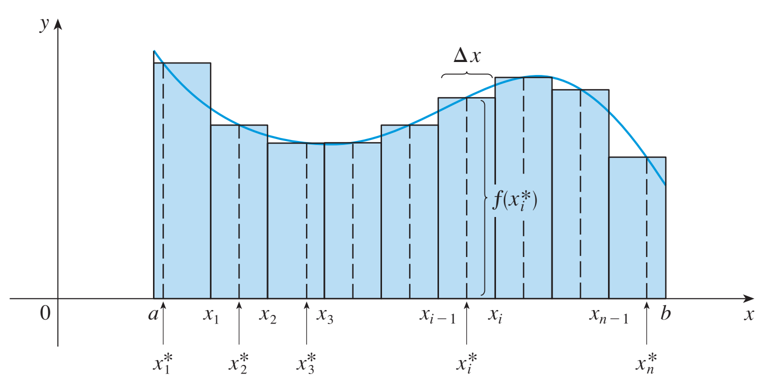


FIGURE 13

This tells us to end with $i = n$.

This tells us to add. $\displaystyle\sum_{i=m}^{n} f(x_i)\,\Delta x$

This tells us to start with $i = m$.

We often use **sigma notation** to write sums with many terms more compactly. For instance,

$$\sum_{i=1}^{n} f(x_i)\,\Delta x = f(x_1)\,\Delta x + f(x_2)\,\Delta x + \cdots + f(x_n)\,\Delta x$$

So the expressions for area in Equations 2, 3, and 4 can be written as follows:

If you need practice with sigma notation, look at the examples and try some of the exercises in Appendix F.

$$A = \lim_{n \to \infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$$

$$A = \lim_{n \to \infty} \sum_{i=1}^{n} f(x_{i-1})\,\Delta x$$

$$A = \lim_{n \to \infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

We can also rewrite Formula 1 in the following way:

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

EXAMPLE 3 | Let A be the area of the region that lies under the graph of $f(x) = e^{-x}$ between $x = 0$ and $x = 2$.

(a) Using right endpoints, find an expression for A as a limit. Do not evaluate the limit.

(b) Estimate the area by taking the sample points to be midpoints and using four subintervals and then ten subintervals.

SOLUTION

(a) Since $a = 0$ and $b = 2$, the width of a subinterval is

$$\Delta x = \frac{2-0}{n} = \frac{2}{n}$$

So $x_1 = 2/n$, $x_2 = 4/n$, $x_3 = 6/n$, $x_i = 2i/n$, and $x_n = 2n/n$. The sum of the areas of the approximating rectangles is

$$\begin{aligned} R_n &= f(x_1)\,\Delta x + f(x_2)\,\Delta x + \cdots + f(x_n)\,\Delta x \\ &= e^{-x_1}\,\Delta x + e^{-x_2}\,\Delta x + \cdots + e^{-x_n}\,\Delta x \\ &= e^{-2/n}\left(\frac{2}{n}\right) + e^{-4/n}\left(\frac{2}{n}\right) + \cdots + e^{-2n/n}\left(\frac{2}{n}\right) \end{aligned}$$

According to Definition 2, the area is

$$A = \lim_{n \to \infty} R_n = \lim_{n \to \infty} \frac{2}{n}\left(e^{-2/n} + e^{-4/n} + e^{-6/n} + \cdots + e^{-2n/n}\right)$$

Using sigma notation we could write

$$A = \lim_{n \to \infty} \frac{2}{n} \sum_{i=1}^{n} e^{-2i/n}$$

It is difficult to evaluate this limit directly by hand, though it isn't hard with the help of a computer algebra system. In Section 5.3 we will be able to find A more easily using a different method.

(b) With $n = 4$ the subintervals of equal width $\Delta x = 0.5$ are $[0, 0.5]$, $[0.5, 1]$, $[1, 1.5]$, and $[1.5, 2]$. The midpoints of these subintervals are $x_1^* = 0.25$, $x_2^* = 0.75$, $x_3^* = 1.25$, and $x_4^* = 1.75$, and the sum of the areas of the four approximating rectangles (see

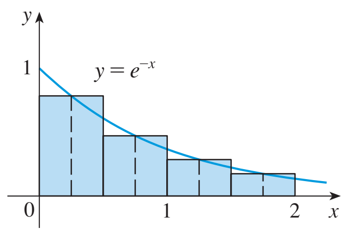


FIGURE 14

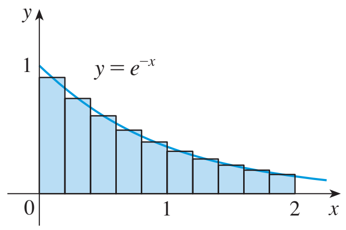


FIGURE 15

Figure 14) is

$$
\begin{aligned}
M_4 &= \sum_{i=1}^{4} f(x_i^*)\,\Delta x \\
&= f(0.25)\,\Delta x + f(0.75)\,\Delta x + f(1.25)\,\Delta x + f(1.75)\,\Delta x \\
&= e^{-0.25}(0.5) + e^{-0.75}(0.5) + e^{-1.25}(0.5) + e^{-1.75}(0.5) \\
&= \tfrac{1}{2}(e^{-0.25} + e^{-0.75} + e^{-1.25} + e^{-1.75}) \approx 0.8557
\end{aligned}
$$

So an estimate for the area is

$$A \approx 0.8557$$

With $n = 10$ the subintervals are $[0, 0.2], [0.2, 0.4], \ldots, [1.8, 2]$ and the midpoints are $x_1^* = 0.1$, $x_2^* = 0.3$, $x_3^* = 0.5, \ldots, x_{10}^* = 1.9$. Thus

$$
\begin{aligned}
A \approx M_{10} &= f(0.1)\,\Delta x + f(0.3)\,\Delta x + f(0.5)\,\Delta x + \cdots + f(1.9)\,\Delta x \\
&= 0.2(e^{-0.1} + e^{-0.3} + e^{-0.5} + \cdots + e^{-1.9}) \approx 0.8632
\end{aligned}
$$

From Figure 15 it appears that this estimate is better than the estimate with $n = 4$. ■

■ The Distance Problem

Now let's consider the *distance problem*: Find the distance traveled by an object during a certain time period if the velocity of the object is known at all times. (In a sense this is the inverse problem of the velocity problem that we discussed in Section 2.3.) If the velocity remains constant, then the distance problem is easy to solve by means of the formula

$$\text{distance} = \text{velocity} \times \text{time}$$

But if the velocity varies, it's not so easy to find the distance traveled. We investigate the problem in the following example.

EXAMPLE 4 | Suppose the odometer on our car is broken and we want to estimate the distance driven over a 30-second time interval. We take speedometer readings every five seconds and record them in the following table:

Time (s)	0	5	10	15	20	25	30
Velocity (mi/h)	17	21	24	29	32	31	28

In order to have the time and the velocity in consistent units, let's convert the velocity readings to feet per second (1 mi/h = 5280/3600 ft/s):

Time (s)	0	5	10	15	20	25	30
Velocity (ft/s)	25	31	35	43	47	45	41

During the first five seconds the velocity doesn't change very much, so we can estimate the distance traveled during that time by assuming that the velocity is constant. If we take the velocity during that time interval to be the initial velocity (25 ft/s), then we obtain the approximate distance traveled during the first five seconds:

$$25 \text{ ft/s} \times 5 \text{ s} = 125 \text{ ft}$$

Similarly, during the second time interval the velocity is approximately constant and we take it to be the velocity when $t = 5$ s. So our estimate for the distance traveled from $t = 5$ s to $t = 10$ s is

$$31 \text{ ft/s} \times 5 \text{ s} = 155 \text{ ft}$$

If we add similar estimates for the other time intervals, we obtain an estimate for the total distance traveled:

$$(25 \times 5) + (31 \times 5) + (35 \times 5) + (43 \times 5) + (47 \times 5) + (45 \times 5) = 1130 \text{ ft}$$

We could just as well have used the velocity at the *end* of each time period instead of the velocity at the beginning as our assumed constant velocity. Then our estimate becomes

$$(31 \times 5) + (35 \times 5) + (43 \times 5) + (47 \times 5) + (45 \times 5) + (41 \times 5) = 1210 \text{ ft}$$

If we had wanted a more accurate estimate, we could have taken velocity readings every two seconds, or even every second. ■

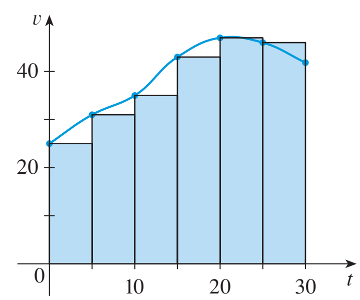


FIGURE 16

Perhaps the calculations in Example 4 remind you of the sums we used earlier to estimate areas. The similarity is explained when we sketch a graph of the velocity function of the car in Figure 16 and draw rectangles whose heights are the initial velocities for each time interval. The area of the first rectangle is $25 \times 5 = 125$, which is also our estimate for the distance traveled in the first five seconds. In fact, the area of each rectangle can be interpreted as a distance because the height represents velocity and the width represents time. The sum of the areas of the rectangles in Figure 16 is $L_6 = 1130$, which is our initial estimate for the total distance traveled.

In general, suppose an object moves with velocity $v = f(t)$, where $a \leqslant t \leqslant b$ and $f(t) \geqslant 0$ (so the object always moves in the positive direction). We take velocity readings at times $t_0 \,(= a), t_1, t_2, \ldots, t_n \,(= b)$ so that the velocity is approximately constant on each subinterval. If these times are equally spaced, then the time between consecutive readings is $\Delta t = (b - a)/n$. During the first time interval the velocity is approximately $f(t_0)$ and so the distance traveled is approximately $f(t_0)\,\Delta t$. Similarly, the distance traveled during the second time interval is about $f(t_1)\,\Delta t$ and the total distance traveled during the time interval $[a, b]$ is approximately

$$f(t_0)\,\Delta t + f(t_1)\,\Delta t + \cdots + f(t_{n-1})\,\Delta t = \sum_{i=1}^{n} f(t_{i-1})\,\Delta t$$

If we use the velocity at right endpoints instead of left endpoints, our estimate for the total distance becomes

$$f(t_1)\,\Delta t + f(t_2)\,\Delta t + \cdots + f(t_n)\,\Delta t = \sum_{i=1}^{n} f(t_i)\,\Delta t$$

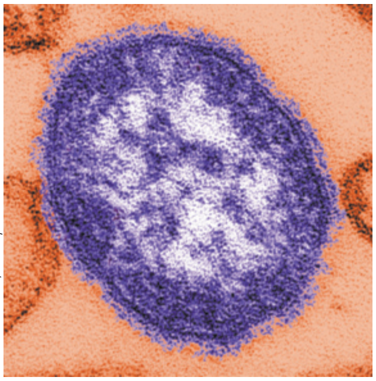
The more frequently we measure the velocity, the more accurate our estimates become, so it seems plausible that the *exact* distance d traveled is the *limit* of such expressions:

$$\textbf{(5)} \qquad d = \lim_{n\to\infty} \sum_{i=1}^{n} f(t_{i-1})\,\Delta t = \lim_{n\to\infty} \sum_{i=1}^{n} f(t_i)\,\Delta t$$

We will see in Section 5.3 that this is indeed true.

Because Equation 5 has the same form as our expressions for area in Equations 2 and 3, it follows that the distance traveled is equal to the area under the graph of the velocity function.

Pathogenesis


Scott Camazine / Alamy

Measles is a highly contagious infection of the respiratory system and is caused by a virus. Despite the fact that more than 80% of the world's population is vaccinated for it, measles remains the fifth leading cause of death worldwide.

In general, the term *pathogenesis* refers to the way a disease originates and develops over time. In the case of measles, the virus enters through the respiratory tract and replicates there before spreading into the bloodstream and then the skin. In a person with no immunity to measles the characteristic rash usually appears about 12 days after infection. The virus reaches a peak density in the blood at about 14 days. The virus level then decreases fairly rapidly over the next few days as a result of the immune response. This progression is reflected in the pathogenesis curve in Figure 17. Notice that the vertical axis is measured in units of number of infected cells per mL of blood plasma.

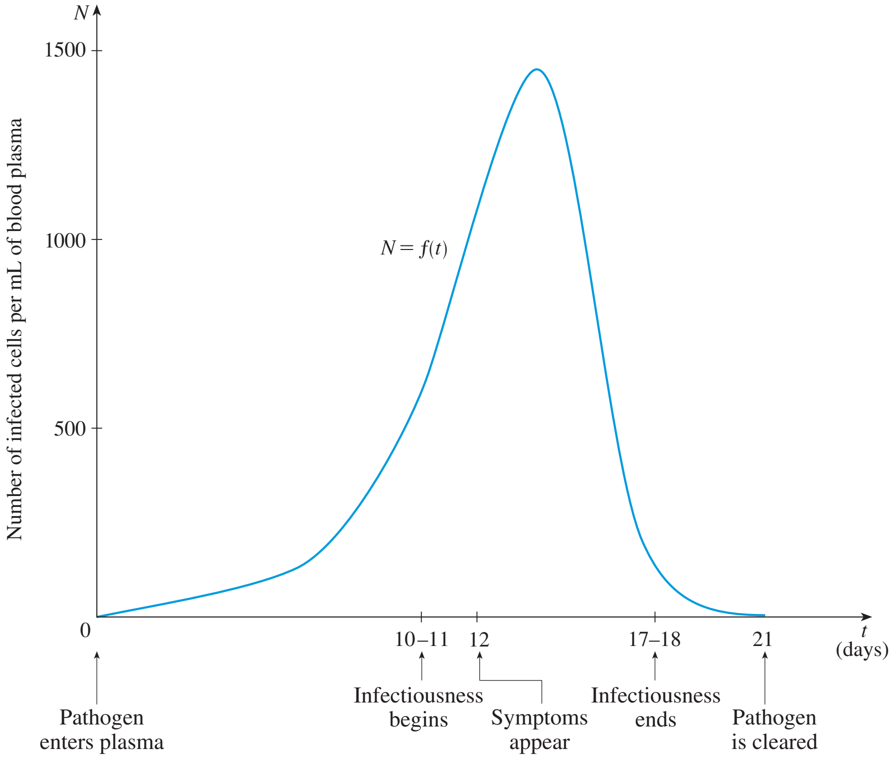


FIGURE 17
Measles pathogenesis curve
Source: J. M. Heffernan et al., "An In-Host Model of Acute Infection: Measles as a Case Study," *Theoretical Population Biology* 73 (2008): 134–47.

Let's denote by f the measles pathogenesis function in Figure 17. Therefore $f(t)$ gives the number of infected cells per mL of plasma on day t. Measles symptoms are thought to develop only after the immune system has been exposed to a threshold "amount of infection." The amount of infection is determined by both the number of infected cells per mL and by the duration over which these cells are exposed to the immune system. If the density of infected cells were constant during infection, then the total

amount of infection would be measured as

$$\text{amount of infection} = \text{density of infected cells} \times \text{time}$$

with the units being (number of cells per mL) $\times$ days. Of course the density is not constant, but we can break the duration of infection into shorter time intervals over which the density changes very little. If each of these shorter time intervals has width Δt, we could add the areas $f(t_i)\,\Delta t$ of the rectangles in Figure 18 and get an approximation to the amount of infection over the first 12 days. Then we take the limit as $\Delta t \to 0$ and the number of rectangles becomes large. By an argument like the one that led to Equation 5, we conclude that the amount of infection needed to stimulate the appearance of symptoms is as follows.

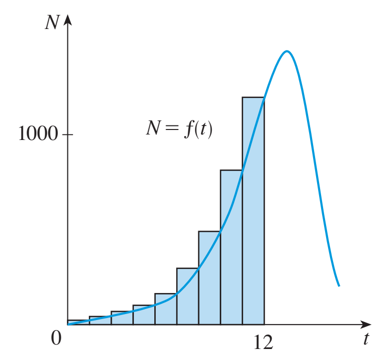


FIGURE 18

> The area under the pathogenesis curve $N = f(t)$ from $t = 0$ to $t = 12$ (shaded in Figure 19) is equal to the total amount of infection needed to develop symptoms.

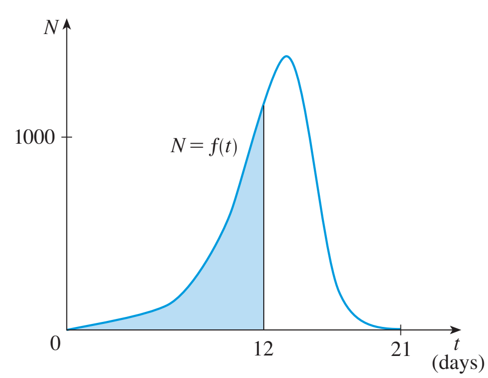


FIGURE 19
Area under pathogenesis curve up to 12 days is the amount of infection needed for symptoms.

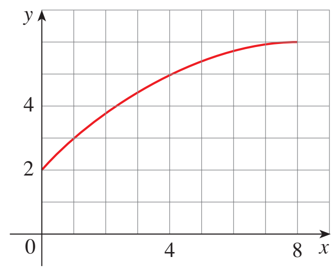
The measles pathogenesis curve has been modeled by the polynomial

$$f(t) = -t(t - 21)(t + 1)$$

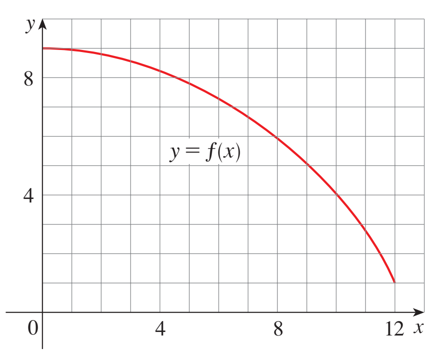
In Exercise 9 you are asked to use this model to estimate the area under the curve $N = f(t)$ and therefore the total amount of infection needed for the patient to be symptomatic.

EXERCISES 5.1

1. (a) By reading values from the given graph of f, use four rectangles to find a lower estimate and an upper estimate for the area under the given graph of f from $x = 0$ to $x = 8$. In each case sketch the rectangles that you use.
(b) Find new estimates using eight rectangles in each case.



2. (a) Use six rectangles to find estimates of each type for the area under the given graph of f from $x = 0$ to $x = 12$.
(i) L_6 (sample points are left endpoints)
(ii) R_6 (sample points are right endpoints)
(iii) M_6 (sample points are midpoints)
(b) Is L_6 an underestimate or overestimate of the true area?
(c) Is R_6 an underestimate or overestimate of the true area?
(d) Which of the numbers L_6, R_6, or M_6 gives the best estimate? Explain.



3. (a) Estimate the area under the graph of $f(x) = \cos x$ from $x = 0$ to $x = \pi/2$ using four approximating rectangles and right endpoints. Sketch the graph and the rectangles. Is your estimate an underestimate or an overestimate?
(b) Repeat part (a) using left endpoints.

4. (a) Estimate the area under the graph of $f(x) = \sqrt{x}$ from $x = 0$ to $x = 4$ using four approximating rectangles and right endpoints. Sketch the graph and the rectangles. Is your estimate an underestimate or an overestimate?
(b) Repeat part (a) using left endpoints.

5. (a) Estimate the area under the graph of $f(x) = 1 + x^2$ from $x = -1$ to $x = 2$ using three rectangles and right endpoints. Then improve your estimate by using six rectangles. Sketch the curve and the approximating rectangles.
(b) Repeat part (a) using left endpoints.
(c) Repeat part (a) using midpoints.
(d) From your sketches in parts (a)–(c), which appears to be the best estimate?

6. (a) Graph the function

$$f(x) = x - 2\ln x \qquad 1 \leqslant x \leqslant 5$$

(b) Estimate the area under the graph of f using four approximating rectangles and taking the sample points to be (i) right endpoints and (ii) midpoints. In each case sketch the curve and the rectangles.
(c) Improve your estimates in part (b) by using eight rectangles.

7. The speed of a runner increased steadily during the first three seconds of a race. Her speed at half-second intervals is given in the table. Find lower and upper estimates for the distance that she traveled during these three seconds.

t (s)	0	0.5	1.0	1.5	2.0	2.5	3.0
v (ft/s)	0	6.2	10.8	14.9	18.1	19.4	20.2

8. Speedometer readings for a motorcycle at 12-second intervals are given in the table.
(a) Estimate the distance traveled by the motorcycle during this time period using the velocities at the beginning of the time intervals.
(b) Give another estimate using the velocities at the end of the time periods.
(c) Are your estimates in parts (a) and (b) upper and lower estimates? Explain.

t (s)	0	12	24	36	48	60
v (ft/s)	30	28	25	22	24	27

9. Measles pathogenesis The function

$$f(t) = -t(t - 21)(t + 1)$$

can be used to model the measles pathogenesis curve in Figures 17 and 19. Suppose symptoms appear after 12 days. Use six subintervals and their midpoints to estimate the total amount of infection needed to develop symptoms.

10. Measles pathogenesis If a patient has had previous exposure to measles, the immune system responds more quickly. This results in a suppression of the level of virus in the plasma during an infection. Suppose that such previous exposure causes the viral density in the plasma to be $\frac{3}{5}$ of that in a patient with no previous exposure. This means that the level of virus in the plasma is given by $\frac{3}{5}f(t)$, where $f(t) = -t(t - 21)(t + 1)$ as in Exercise 9.

(a) Use subintervals of width 2 days and their midpoints to estimate the total amount of infection at $t = 12$ days.

(b) If 7848 cells per mL $\times$ days is the total amount of infection required to develop symptoms, use subintervals of width 2 days and their midpoints to estimate the day when symptoms will appear.

11. SARS incidence The table shows the number of people per day who died from SARS in Singapore at two-week intervals beginning on March 1, 2003.

Date	Deaths per day	Date	Deaths per day
March 1	0.0079	April 26	0.5620
March 15	0.0638	May 10	0.4630
March 29	0.1944	May 24	0.2897
April 12	0.4435		

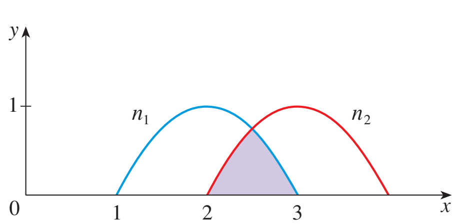
Estimate the number of people who died of SARS in Singapore between March 1 and May 24, 2003, using both left endpoints and right endpoints.

Source: Adapted from A. Gumel et al., "Modelling Strategies for Controlling SARS Outbreaks," *Proceedings of the Royal Society of London: Series B* 271 (2004): 2223–32.

12. Niche overlap The extent to which species compete for resources is often measured by the *niche overlap*. If the horizontal axis represents a continuum of different resource types (for example, seed sizes for certain bird species), then a plot of the degree of preference for these resources is called a *species' niche*. The degree of overlap of two species' niches is then a measure of the extent to which they compete for resources. The niche overlap for a species is the fraction of the area under its preference curve that is also under the other species' curve. The niches displayed in the figure are given by

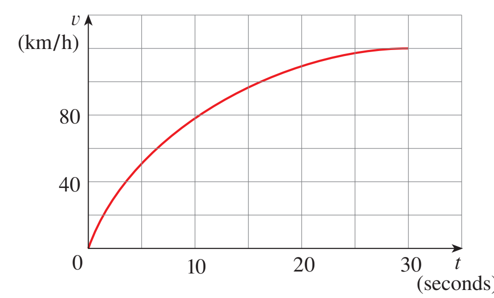
$$n_1(x) = (x - 1)(3 - x) \qquad 1 \leqslant x \leqslant 3$$
$$n_2(x) = (x - 2)(4 - x) \qquad 2 \leqslant x \leqslant 4$$

Estimate the niche overlap for species 1 using midpoints. (Choose the number of subintervals yourself.)



13. The velocity graph of a braking car is shown. Use it to estimate the distance traveled by the car while the brakes are applied.

14. The velocity graph of a car accelerating from rest to a speed of 120 km/h over a period of 30 seconds is shown. Estimate the distance traveled during this period.



15–17 Use Definition 2 to find an expression for the area under the graph of f as a limit. Do not evaluate the limit.

15. $f(x) = \dfrac{2x}{x^2 + 1}, \quad 1 \leqslant x \leqslant 3$

16. $f(x) = x^2 + \sqrt{1 + 2x}, \quad 4 \leqslant x \leqslant 7$

17. $f(x) = x \cos x, \quad 0 \leqslant x \leqslant \pi/2$

18. (a) Use Definition 2 to find an expression for the area under the curve $y = x^3$ from 0 to 1 as a limit.

(b) The following formula for the sum of the cubes of the first n integers is proved in Appendix E. Use it to evaluate the limit in part (a).

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

19. Let A be the area under the graph of an increasing continuous function f from a to b, and let L_n and R_n be the approximations to A with n subintervals using left and right endpoints, respectively.

(a) How are A, L_n, and R_n related?

(b) Show that

$$R_n - L_n = \frac{b - a}{n}[f(b) - f(a)]$$

(c) Deduce that

$$R_n - A < \frac{b-a}{n}[f(b) - f(a)]$$

20. If A is the area under the curve $y = e^x$ from 1 to 3, use Exercise 19 to find a value of n such that $R_n - A < 0.0001$.

CAS **21.** (a) Express the area under the curve $y = x^5$ from 0 to 2 as a limit.
(b) Use a computer algebra system to find the sum in your expression from part (a).
(c) Evaluate the limit in part (a).

CAS **22.** Find the exact area of the region under the graph of $y = e^{-x}$ from 0 to 2 by using a computer algebra system to evaluate the sum and then the limit in Example 3(a). Compare your answer with the estimate obtained in Example 3(b).

CAS **23.** Find the exact area under the cosine curve $y = \cos x$ from $x = 0$ to $x = b$, where $0 \leqslant b \leqslant \pi/2$. (Use a computer algebra system both to evaluate the sum and compute the limit.) In particular, what is the area if $b = \pi/2$?

24. (a) Let A_n be the area of a polygon with n equal sides inscribed in a circle with radius r. By dividing the polygon into n congruent triangles with central angle $2\pi/n$, show that

$$A_n = \tfrac{1}{2}nr^2 \sin\left(\frac{2\pi}{n}\right)$$

(b) Show that $\lim_{n\to\infty} A_n = \pi r^2$. [*Hint:* Use Equation 2.4.6 on page 133.]

5.2 The Definite Integral

We saw in Section 5.1 that a limit of the form

$$\textbf{(1)} \qquad \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x = \lim_{n\to\infty} [f(x_1^*)\,\Delta x + f(x_2^*)\,\Delta x + \cdots + f(x_n^*)\,\Delta x]$$

arises when we compute an area. We also saw that it arises when we try to find the distance traveled by an object or the total amount of infection needed to show measles symptoms. It turns out that this same type of limit occurs in a wide variety of situations even when f is not necessarily a positive function. In Chapter 6 we will see that limits of the form (1) also arise in finding volumes of tumors, cardiac output, blood flow, and many other quantities. We therefore give this type of limit a special name and notation.

A precise definition of this type of limit is given in Appendix D.

(2) Definition of a Definite Integral If f is a function defined for $a \leqslant x \leqslant b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$. We let $x_0\,(= a), x_1, x_2, \ldots, x_n\,(= b)$ be the endpoints of these subintervals and we let $x_1^*, x_2^*, \ldots, x_n^*$ be any **sample points** in these subintervals, so x_i^* lies in the ith subinterval $[x_{i-1}, x_i]$. Then the **definite integral of *f* from *a* to *b*** is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

provided that this limit exists. If it does exist, we say that f is **integrable** on $[a, b]$.

NOTE 1 The symbol $\int$ was introduced by Leibniz and is called an **integral sign**. It is an elongated S and was chosen because an integral is a limit of sums. In the notation $\int_a^b f(x)\,dx$, $f(x)$ is called the **integrand** and a and b are called the **limits of integration**; a is the **lower limit** and b is the **upper limit**. For now, the symbol dx has no meaning by itself; $\int_a^b f(x)\,dx$ is all one symbol. The dx simply indicates that the independent variable is x. The procedure of calculating an integral is called **integration**.

Riemann

Bernhard Riemann received his Ph.D. under the direction of the legendary Gauss at the University of Göttingen and remained there to teach. Gauss, who was not in the habit of praising other mathematicians, spoke of Riemann's "creative, active, truly mathematical mind and gloriously fertile originality." The definition (2) of an integral that we use is due to Riemann. He also made major contributions to the theory of functions of a complex variable, mathematical physics, number theory, and the foundations of geometry. Riemann's broad concept of space and geometry turned out to be the right setting, 50 years later, for Einstein's general relativity theory. Riemann's health was poor throughout his life, and he died of tuberculosis at the age of 39.

NOTE 2 The definite integral $\int_a^b f(x)\,dx$ is a number; it does not depend on x. In fact, we could use any letter in place of x without changing the value of the integral:

$$\int_a^b f(x)\,dx = \int_a^b f(t)\,dt = \int_a^b f(r)\,dr$$

NOTE 3 The sum

$$\sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

that occurs in Definition 2 is called a **Riemann sum** after the German mathematician Bernhard Riemann (1826–1866). So Definition 2 says that the definite integral of an integrable function can be approximated to within any desired degree of accuracy by a Riemann sum.

We know that if f happens to be positive, then the Riemann sum can be interpreted as a sum of areas of approximating rectangles (see Figure 1). By comparing Definition 2 with the definition of area in Section 5.1, we see that the definite integral $\int_a^b f(x)\,dx$ can be interpreted as the area under the curve $y = f(x)$ from a to b. (See Figure 2.)

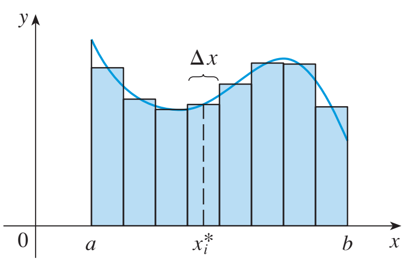


FIGURE 1
If $f(x) \geqslant 0$, the Riemann sum $\Sigma\, f(x_i^*)\,\Delta x$ is the sum of areas of rectangles.

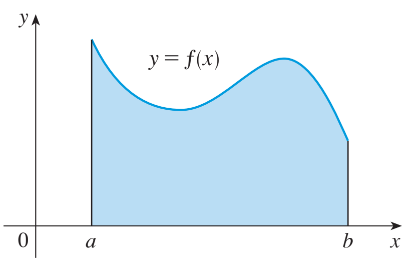


FIGURE 2
If $f(x) \geqslant 0$, the integral $\int_a^b f(x)\,dx$ is the area under the curve $y = f(x)$ from a to b.

If f takes on both positive and negative values, as in Figure 3, then the Riemann sum is the sum of the areas of the rectangles that lie above the x-axis and the *negatives* of the areas of the rectangles that lie below the x-axis (the areas of the blue rectangles *minus* the areas of the gold rectangles). When we take the limit of such Riemann sums, we get the situation illustrated in Figure 4. A definite integral can be interpreted as a **net area**, that is, a difference of areas:

$$\int_a^b f(x)\,dx = A_1 - A_2$$

where A_1 is the area of the region above the x-axis and below the graph of f, and A_2 is the area of the region below the x-axis and above the graph of f.

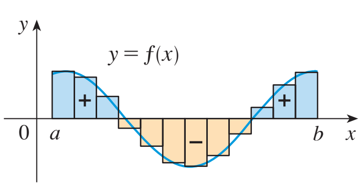


FIGURE 3
$\Sigma\, f(x_i^*)\,\Delta x$ is an approximation to the net area.

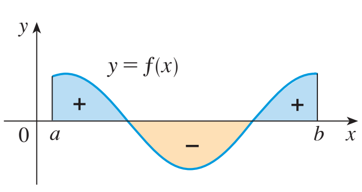


FIGURE 4
$\int_a^b f(x)\,dx$ is the net area.

NOTE 4 Although we have defined $\int_a^b f(x)\,dx$ by dividing $[a, b]$ into subintervals of equal width, there are situations in which it is advantageous to work with subintervals of unequal width. For instance, if in a biological experiment data are collected at times that are not equally spaced, then we can still estimate the area under a curve. If the subinterval widths are $\Delta x_1, \Delta x_2, \ldots, \Delta x_n$, we have to ensure that all these widths approach 0 in the limiting process. This happens if the largest width, $\max \Delta x_i$, approaches 0. So in this case the definition of a definite integral becomes

$$\int_a^b f(x)\,dx = \lim_{\max \Delta x_i \to 0} \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i$$

NOTE 5 We have defined the definite integral for an integrable function, but not all functions are integrable. The following theorem shows that the most commonly occurring functions are in fact integrable. It is proved in more advanced courses.

(3) Theorem If f is continuous on $[a, b]$, or if f has only a finite number of jump discontinuities, then f is integrable on $[a, b]$; that is, the definite integral $\int_a^b f(x)\,dx$ exists.

If f is integrable on $[a, b]$, then the limit in Definition 2 exists and gives the same value no matter how we choose the sample points x_i^*. To simplify the calculation of the integral we often take the sample points to be right endpoints. Then $x_i^* = x_i$ and the definition of an integral simplifies as follows.

(4) Theorem If f is integrable on $[a, b]$, then

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$$

where $\quad \Delta x = \dfrac{b-a}{n} \quad$ and $\quad x_i = a + i\,\Delta x$

EXAMPLE 1 | Express

$$\lim_{n\to\infty} \sum_{i=1}^{n} (x_i^3 + x_i \sin x_i)\,\Delta x$$

as an integral on the interval $[0, \pi]$.

SOLUTION Comparing the given limit with the limit in Theorem 4, we see that they will be identical if we choose $f(x) = x^3 + x \sin x$. We are given that $a = 0$ and $b = \pi$. Therefore, by Theorem 4, we have

$$\lim_{n\to\infty} \sum_{i=1}^{n} (x_i^3 + x_i \sin x_i)\,\Delta x = \int_0^\pi (x^3 + x \sin x)\,dx$$ ■

Later, when we apply the definite integral in biological contexts, it will be important to recognize limits of sums as integrals, as we did in Example 1. When Leibniz chose the notation for an integral, he chose the ingredients as reminders of the limiting process. In general, when we write

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x = \int_a^b f(x)\,dx$$

we replace $\lim \Sigma$ by $\int$, x_i^* by x, and Δx by dx.

■ Calculating Integrals

When we use a limit to evaluate a definite integral, we need to know how to work with sums. The following three equations give formulas for sums of powers of positive integers. Equation 5 may be familiar to you from a course in algebra. Equations 6 and 7 were discussed in Section 5.1 and are proved in Appendix F.

(5) $$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

(6) $$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

(7) $$\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$

The remaining formulas are simple rules for working with sigma notation:

Formulas 8–11 are proved by writing out each side in expanded form. The left side of Equation 9 is

$$ca_1 + ca_2 + \cdots + ca_n$$

The right side is

$$c(a_1 + a_2 + \cdots + a_n)$$

These are equal by the distributive property. The other formulas are discussed in Appendix F.

(8) $$\sum_{i=1}^{n} c = nc$$

(9) $$\sum_{i=1}^{n} ca_i = c\sum_{i=1}^{n} a_i$$

(10) $$\sum_{i=1}^{n} (a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$$

(11) $$\sum_{i=1}^{n} (a_i - b_i) = \sum_{i=1}^{n} a_i - \sum_{i=1}^{n} b_i$$

EXAMPLE 2

(a) Evaluate the Riemann sum for $f(x) = x^3 - 6x$, taking the sample points to be right endpoints and $a = 0$, $b = 3$, and $n = 6$.

(b) Evaluate $\int_0^3 (x^3 - 6x)\,dx$.

SOLUTION

(a) With $n = 6$ the interval width is

$$\Delta x = \frac{b-a}{n} = \frac{3-0}{6} = \frac{1}{2}$$

and the right endpoints are $x_1 = 0.5$, $x_2 = 1.0$, $x_3 = 1.5$, $x_4 = 2.0$, $x_5 = 2.5$, and $x_6 = 3.0$. So the Riemann sum is

$$\begin{aligned} R_6 &= \sum_{i=1}^{6} f(x_i)\,\Delta x \\ &= f(0.5)\,\Delta x + f(1.0)\,\Delta x + f(1.5)\,\Delta x + f(2.0)\,\Delta x + f(2.5)\,\Delta x + f(3.0)\,\Delta x \\ &= \tfrac{1}{2}(-2.875 - 5 - 5.625 - 4 + 0.625 + 9) \\ &= -3.9375 \end{aligned}$$

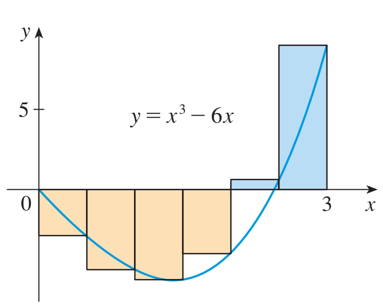


FIGURE 5

Notice that f is not a positive function and so the Riemann sum does not represent a sum of areas of rectangles. But it does represent the sum of the areas of the blue rectangles (above the x-axis) minus the sum of the areas of the gold rectangles (below the x-axis) in Figure 5.

(b) With n subintervals we have

$$\Delta x = \frac{b-a}{n} = \frac{3}{n}$$

Thus $x_0 = 0$, $x_1 = 3/n$, $x_2 = 6/n$, $x_3 = 9/n$, and, in general, $x_i = 3i/n$. Since we are using right endpoints, we can use Theorem 4:

In the sum, n is a constant (unlike i), so we can move $3/n$ in front of the Σ sign.

$$\begin{aligned}
\int_0^3 (x^3 - 6x)\,dx &= \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(\frac{3i}{n}\right)\frac{3}{n} \\
&= \lim_{n\to\infty} \frac{3}{n} \sum_{i=1}^{n} \left[\left(\frac{3i}{n}\right)^3 - 6\left(\frac{3i}{n}\right)\right] && \text{(Equation 9 with } c = 3/n\text{)} \\
&= \lim_{n\to\infty} \frac{3}{n} \sum_{i=1}^{n} \left[\frac{27}{n^3} i^3 - \frac{18}{n} i\right] \\
&= \lim_{n\to\infty} \left[\frac{81}{n^4} \sum_{i=1}^{n} i^3 - \frac{54}{n^2} \sum_{i=1}^{n} i\right] && \text{(Equations 11 and 9)} \\
&= \lim_{n\to\infty} \left\{\frac{81}{n^4}\left[\frac{n(n+1)}{2}\right]^2 - \frac{54}{n^2}\frac{n(n+1)}{2}\right\} && \text{(Equations 7 and 5)} \\
&= \lim_{n\to\infty} \left[\frac{81}{4}\left(1 + \frac{1}{n}\right)^2 - 27\left(1 + \frac{1}{n}\right)\right] \\
&= \frac{81}{4} - 27 = -\frac{27}{4} = -6.75
\end{aligned}$$

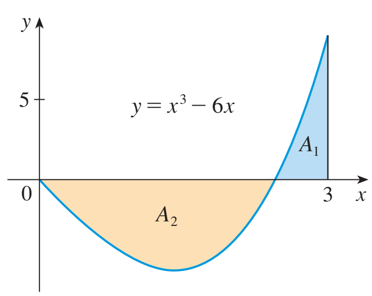


FIGURE 6
$\int_0^3 (x^3 - 6x)\,dx = A_1 - A_2 = -6.75$

This integral can't be interpreted as an area because f takes on both positive and negative values. But it can be interpreted as the difference of areas $A_1 - A_2$, where A_1 and A_2 are shown in Figure 6.

Figure 7 illustrates the calculation by showing the positive and negative terms in the right Riemann sum R_n for $n = 40$. The values in the table show the Riemann sums approaching the exact value of the integral, -6.75, as $n \to \infty$.

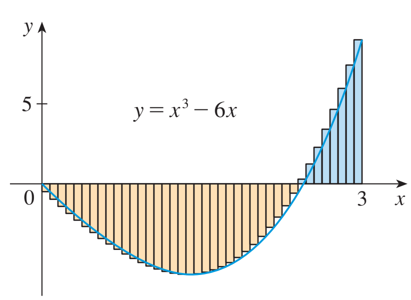


FIGURE 7
$R_{40} \approx -6.3998$

n	R_n
40	−6.3998
100	−6.6130
500	−6.7229
1000	−6.7365
5000	−6.7473

A much simpler method for evaluating the integral in Example 2 will be given in Section 5.3 after we have proved the Evaluation Theorem.

EXAMPLE 3 | Evaluate the following integrals by interpreting each in terms of areas.

(a) $\int_0^1 \sqrt{1 - x^2}\, dx$ (b) $\int_0^3 (x - 1)\, dx$

SOLUTION

(a) Since $f(x) = \sqrt{1 - x^2} \geqslant 0$, we can interpret this integral as the area under the curve $y = \sqrt{1 - x^2}$ from 0 to 1. But, since $y^2 = 1 - x^2$, we get $x^2 + y^2 = 1$, which shows that the graph of f is the quarter-circle with radius 1 in Figure 8. Therefore

$$\int_0^1 \sqrt{1 - x^2}\, dx = \tfrac{1}{4}\pi(1)^2 = \frac{\pi}{4}$$

(b) The graph of $y = x - 1$ is the line with slope 1 shown in Figure 9. We compute the integral as the difference of the areas of the two triangles:

$$\int_0^3 (x - 1)\, dx = A_1 - A_2 = \tfrac{1}{2}(2 \cdot 2) - \tfrac{1}{2}(1 \cdot 1) = 1.5$$

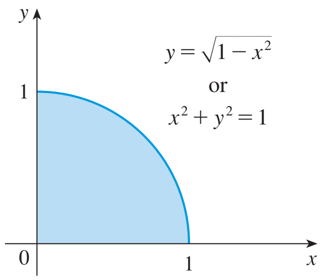


FIGURE 8

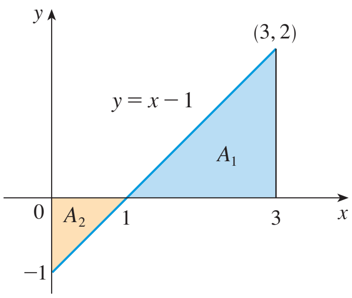


FIGURE 9

The Midpoint Rule

We often choose the sample point x_i^* to be the right endpoint of the ith subinterval because it is convenient for computing the limit. But if the purpose is to find an *approximation* to an integral, it is usually better to choose x_i^* to be the midpoint of the interval, which we denote by $\bar{x}_i$. Any Riemann sum is an approximation to an integral, but if we use midpoints we get the following approximation.

TEC Module 5.2 shows how the Midpoint Rule estimates improve as n increases.

Midpoint Rule

$$\int_a^b f(x)\, dx \approx \sum_{i=1}^{n} f(\bar{x}_i)\, \Delta x = \Delta x\, [f(\bar{x}_1) + \cdots + f(\bar{x}_n)]$$

where $$\Delta x = \frac{b - a}{n}$$

and $$\bar{x}_i = \tfrac{1}{2}(x_{i-1} + x_i) = \text{midpoint of } [x_{i-1}, x_i]$$

EXAMPLE 4 | Use the Midpoint Rule with $n = 5$ to approximate $\int_1^2 \frac{1}{x}\, dx$.

SOLUTION The endpoints of the five subintervals are 1, 1.2, 1.4, 1.6, 1.8, and 2.0, so the midpoints are 1.1, 1.3, 1.5, 1.7, and 1.9. The width of the subintervals is

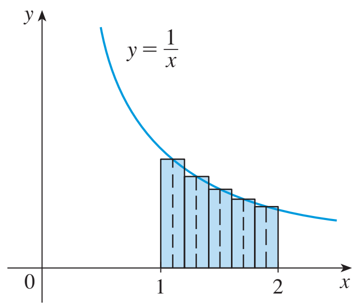


FIGURE 10

$\Delta x = (2 - 1)/5 = \frac{1}{5}$, so the Midpoint Rule gives

$$\int_1^2 \frac{1}{x}\,dx \approx \Delta x\,[f(1.1) + f(1.3) + f(1.5) + f(1.7) + f(1.9)]$$

$$= \frac{1}{5}\left(\frac{1}{1.1} + \frac{1}{1.3} + \frac{1}{1.5} + \frac{1}{1.7} + \frac{1}{1.9}\right)$$

$$\approx 0.691908$$

Since $f(x) = 1/x > 0$ for $1 \leqslant x \leqslant 2$, the integral represents an area, and the approximation given by the Midpoint Rule is the sum of the areas of the rectangles shown in Figure 10. ■

If we apply the Midpoint Rule to the integral in Example 2, we get the picture in Figure 11. The approximation $M_{40} \approx -6.7563$ is much closer to the true value -6.75 than the right endpoint approximation, $R_{40} \approx -6.3998$, shown in Figure 7.

TEC In Visual 5.2 you can compare left, right, and midpoint approximations to the integral in Example 2 for different values of n.

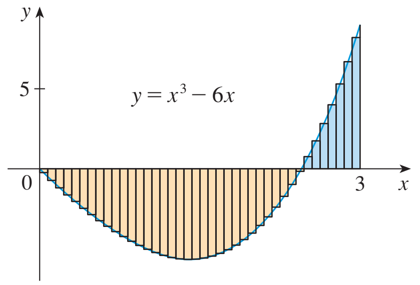


FIGURE 11
$M_{40} \approx -6.7563$

EXAMPLE 5 | Aspirin pharmacokinetics In a study[1] of the effects of low-dose aspirin, 10 young males were given a single 80-mg dose of ASA (acetylsalicylic acid) on three separate days. Blood samples were obtained for 24 hours after each dose and peak plasma ASA levels of about 1 μg/mL were reached within 30 minutes. After the results for the 10 volunteers were averaged, the concentration of ASA in the bloodstream was modeled by the function

$$C(t) = 32t^2e^{-4.2t}$$

where t is measured in hours and C is measured in μg/mL. The graph of C is shown in Figure 12. Use the Midpoint Rule with 10 subintervals to estimate the value of the integral $\int_0^2 C(t)\,dt$. What are the units?

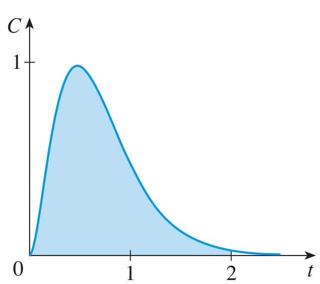


FIGURE 12
Area under the ASA concentration function

SOLUTION If we divide the interval [0, 2] into 10 subintervals, then the midpoints are 0.1, 0.3, 0.5, . . . , 1.7, 1.9 and the width of each subinterval is $\Delta t = 0.2$. Using the Midpoint Rule, we have

$$\int_0^2 C(t)\,dt \approx \Delta t[f(0.1) + f(0.3) + f(0.5) + \cdots + f(1.9)]$$

$$\approx 0.2[0.2103 + 0.8169 + 0.9797 + 0.8289 + 0.5916$$

$$+ 0.3815 + 0.2300 + 0.1322 + 0.0733 + 0.0395$$

$$= 0.8568$$

1. I. H. Benedek et al., "Variability in the Pharmacokinetics and Pharmacodynamics of Low Dose Aspirin in Healthy Male Volunteers," *Journal of Clinical Pharmacology* 35 (1995): 1181–86.

The units for C are micrograms per milliliter (μg/mL) and the units for t are hours, so the units for the integral are micrograms per milliliter times hours:

$$\int_0^2 C(t)\,dt \approx 0.8568\ (\mu\text{g/mL})\cdot\text{h}$$

How would we interpret biologically the integral we computed in Example 5? In the biological literature it is denoted simply by AUC (area under the curve), but what does that mean in this context? For a drug to work, it needs to be "available" to interact with the target tissue (or target pathogen if the drug is an antibiotic). Availability can be increased by increasing the concentration or by increasing the time the drug lingers before it is cleared through metabolism. AUC is a combined measure of these, and therefore is a composite measure of the overall "availability."

Properties of the Definite Integral

When we defined the definite integral $\int_a^b f(x)\,dx$, we implicitly assumed that $a < b$. But the definition as a limit of Riemann sums makes sense even if $a > b$. Notice that if we reverse a and b, then Δx changes from $(b-a)/n$ to $(a-b)/n$. Therefore

$$\boxed{\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx}$$

If $a = b$, then $\Delta x = 0$ and so

$$\boxed{\int_a^a f(x)\,dx = 0}$$

We now develop some basic properties of integrals that will help us to evaluate integrals in a simple manner. We assume that f and g are continuous functions.

Properties of the Integral

1. $\displaystyle\int_a^b c\,dx = c(b-a)$, where c is any constant

2. $\displaystyle\int_a^b [f(x) + g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$

3. $\displaystyle\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$, where c is any constant

4. $\displaystyle\int_a^b [f(x) - g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$

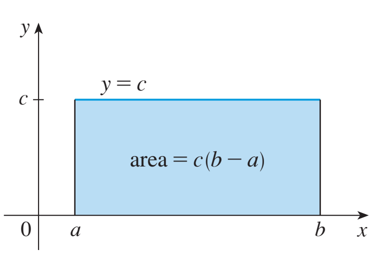


FIGURE 13

$\int_a^b c\,dx = c(b-a)$

Property 1 says that the integral of a constant function $f(x) = c$ is the constant times the length of the interval. If $c > 0$ and $a < b$, this is to be expected because $c(b-a)$ is the area of the shaded rectangle in Figure 13.

Property 2 says that the integral of a sum is the sum of the integrals. For positive functions it says that the area under $f + g$ is the area under f plus the area under g. Figure 14 helps us understand why this is true: In view of how graphical addition works, the corresponding vertical line segments have equal height.

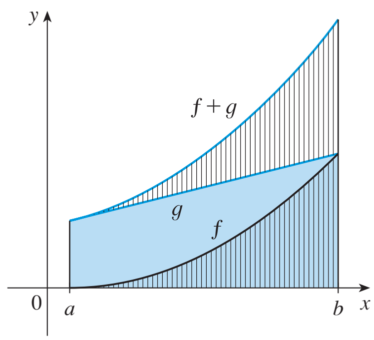


FIGURE 14

$$\int_a^b [f(x) + g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

In general, Property 2 follows from Theorem 4 and the fact that the limit of a sum is the sum of the limits:

$$\begin{aligned}\int_a^b [f(x) + g(x)]\,dx &= \lim_{n\to\infty} \sum_{i=1}^{n} [f(x_i) + g(x_i)]\,\Delta x \\ &= \lim_{n\to\infty} \left[\sum_{i=1}^{n} f(x_i)\,\Delta x + \sum_{i=1}^{n} g(x_i)\,\Delta x\right] \\ &= \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x + \lim_{n\to\infty} \sum_{i=1}^{n} g(x_i)\,\Delta x \\ &= \int_a^b f(x)\,dx + \int_a^b g(x)\,dx\end{aligned}$$

Property 3 seems intuitively reasonable because we know that multiplying a function by a positive number c stretches or shrinks its graph vertically by a factor of c. So it stretches or shrinks each approximating rectangle by a factor c and therefore it has the effect of multiplying the area by c.

Property 3 can be proved in a similar manner and says that the integral of a constant times a function is the constant times the integral of the function. In other words, a constant (but *only* a constant) can be taken in front of an integral sign. Property 4 is proved by writing $f - g = f + (-g)$ and using Properties 2 and 3 with $c = -1$.

EXAMPLE 6 | Use the properties of integrals to evaluate $\int_0^1 (4 + 3x^2)\,dx$.

SOLUTION Using Properties 2 and 3 of integrals, we have

$$\int_0^1 (4 + 3x^2)\,dx = \int_0^1 4\,dx + \int_0^1 3x^2\,dx = \int_0^1 4\,dx + 3\int_0^1 x^2\,dx$$

We know from Property 1 that

$$\int_0^1 4\,dx = 4(1 - 0) = 4$$

and we found in Example 5.1.2 that $\int_0^1 x^2\,dx = \frac{1}{3}$. So

$$\begin{aligned}\int_0^1 (4 + 3x^2)\,dx &= \int_0^1 4\,dx + 3\int_0^1 x^2\,dx \\ &= 4 + 3 \cdot \tfrac{1}{3} = 5\end{aligned}$$

■

The next property tells us how to combine integrals of the same function over adjacent intervals:

5. $$\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx$$

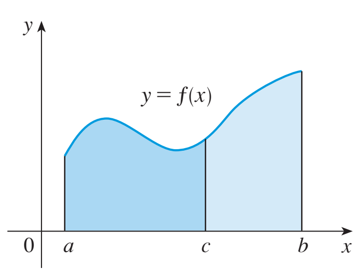


FIGURE 15

This is not easy to prove in general, but for the case where $f(x) \geqslant 0$ and $a < c < b$ Property 5 can be seen from the geometric interpretation in Figure 15: The area under $y = f(x)$ from a to c plus the area from c to b is equal to the total area from a to b.

EXAMPLE 7 | If it is known that $\int_0^{10} f(x)\,dx = 17$ and $\int_0^8 f(x)\,dx = 12$, find $\int_8^{10} f(x)\,dx$.

SOLUTION By Property 5, we have

$$\int_0^8 f(x)\,dx + \int_8^{10} f(x)\,dx = \int_0^{10} f(x)\,dx$$

so
$$\int_8^{10} f(x)\,dx = \int_0^{10} f(x)\,dx - \int_0^8 f(x)\,dx = 17 - 12 = 5$$
■

Properties 1–5 are true whether $a < b$, $a = b$, or $a > b$. The following properties, in which we compare sizes of functions and sizes of integrals, are true only if $a \leqslant b$. Again we assume that the functions are continuous.

Comparison Properties of the Integral

6. If $f(x) \geqslant 0$ for $a \leqslant x \leqslant b$, then $\int_a^b f(x)\,dx \geqslant 0$.

7. If $f(x) \geqslant g(x)$ for $a \leqslant x \leqslant b$, then $\int_a^b f(x)\,dx \geqslant \int_a^b g(x)\,dx$.

8. If $m \leqslant f(x) \leqslant M$ for $a \leqslant x \leqslant b$, then

$$m(b - a) \leqslant \int_a^b f(x)\,dx \leqslant M(b - a)$$

Because we are assuming that f and g are continuous, the inequalities $\geqslant$ and $\leqslant$ can be replaced by the strict inequalities $>$ and $<$.

If $f(x) \geqslant 0$, then $\int_a^b f(x)\,dx$ represents the area under the graph of f, so the geometric interpretation of Property 6 is simply that areas are positive. (It also follows directly from the definition because all the quantities involved are positive.) Property 7 says that a bigger function has a bigger integral. It follows from Properties 6 and 4 because $f - g \geqslant 0$.

Property 8 is illustrated by Figure 16 for the case where $f(x) \geqslant 0$. If f is continuous we could take m and M to be the absolute minimum and maximum values of f on the interval $[a, b]$. In this case Property 8 says that the area under the graph of f is greater than the area of the rectangle with height m and less than the area of the rectangle with height M.

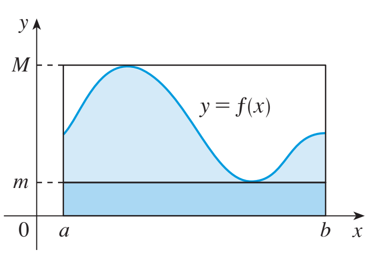


FIGURE 16

PROOF OF PROPERTY 8 Since $m \leqslant f(x) \leqslant M$, Property 7 gives

$$\int_a^b m\,dx \leqslant \int_a^b f(x)\,dx \leqslant \int_a^b M\,dx$$

Using Property 1 to evaluate the integrals on the left and right sides, we obtain

$$m(b - a) \leqslant \int_a^b f(x)\,dx \leqslant M(b - a)$$
■

EXAMPLE 8 | **Measles pathogenesis** In Exercise 5.1.10 we were told that if a patient has had previous exposure to measles, then the level of virus in the plasma during an infection is suppressed (due to stronger immunity). The threshold amount of infection required to develop symptoms is 7848 infected cells per mL $\times$ days. Suppose you and your friend Bob are both infected with measles at the same time but Bob has stronger immunity than you (so that your pathogenesis curve is always at least as high as his). Use Property 7 to show that, if Bob starts to display symptoms on day T, then you must necessarily also display symptoms by this day.

SOLUTION Let's use $f(t)$ and $g(t)$ to be the pathogenesis curves for you and Bob. Because Bob has stronger immunity than you, we know that $f(t) \geqslant g(t)$ at all times (your pathogenesis curve is always at least as high as his). Now if Bob starts to display symptoms on day T, then $\int_0^T g(t)\,dt = 7848$. Furthermore, after T days the level of infection experienced by you will be $\int_0^T f(t)\,dt$. Using Property 7, we see that

$$\int_0^T f(t)\,dt \geqslant \int_0^T g(t)\,dt$$

Therefore we have that $\int_0^T f(t)\,dt \geqslant 7848$, meaning that by day T you will also have reached the threshold amount of infection required to display symptoms. ■

As the next example shows, Property 8 is useful when all we want is a rough estimate of the size of an integral without going to the bother of using the Midpoint Rule.

EXAMPLE 9 | Use Property 8 to estimate $\int_0^1 e^{-x^2}\,dx$.

SOLUTION Because $f(x) = e^{-x^2}$ is a decreasing function on $[0, 1]$, its absolute maximum value is $M = f(0) = 1$ and its absolute minimum value is $m = f(1) = e^{-1}$. Thus, by Property 8,

$$e^{-1}(1 - 0) \leqslant \int_0^1 e^{-x^2}\,dx \leqslant 1(1 - 0)$$

or

$$e^{-1} \leqslant \int_0^1 e^{-x^2}\,dx \leqslant 1$$

Since $e^{-1} \approx 0.3679$, we can write

$$0.367 \leqslant \int_0^1 e^{-x^2}\,dx \leqslant 1$$ ■

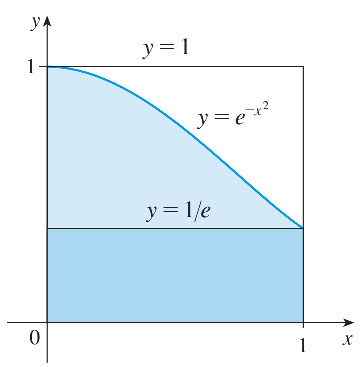


FIGURE 17

The result of Example 9 is illustrated in Figure 17. The integral is greater than the area of the lower rectangle and less than the area of the square.

EXERCISES 5.2

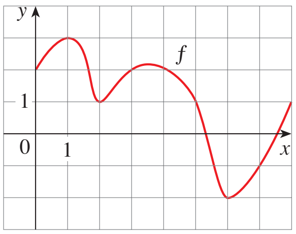
1. Evaluate the Riemann sum for $f(x) = 3 - \frac{1}{2}x$, $2 \leqslant x \leqslant 14$, with six subintervals, taking the sample points to be left endpoints. Explain, with the aid of a diagram, what the Riemann sum represents.

2. If $f(x) = x^2 - 2x$, $0 \leqslant x \leqslant 3$, evaluate the Riemann sum with $n = 6$, taking the sample points to be right endpoints. What does the Riemann sum represent? Illustrate with a diagram.

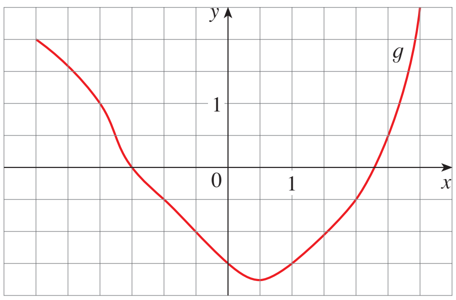
3. If $f(x) = e^x - 2$, $0 \leqslant x \leqslant 2$, find the Riemann sum with $n = 4$ correct to six decimal places, taking the sample points to be midpoints. What does the Riemann sum represent? Illustrate with a diagram.

4. (a) Find the Riemann sum for $f(x) = \sin x$, $0 \leqslant x \leqslant 3\pi/2$, with six terms, taking the sample points to be right endpoints. (Give your answer correct to six decimal places.) Explain what the Riemann sum represents with the aid of a sketch.
   (b) Repeat part (a) with midpoints as the sample points.

5. The graph of a function f is given. Estimate $\int_0^8 f(x)\,dx$ using four subintervals with (a) right endpoints, (b) left endpoints, and (c) midpoints.



6. The graph of g is shown. Estimate $\int_{-3}^{3} g(x)\,dx$ with six subintervals using (a) right endpoints, (b) left endpoints, and (c) midpoints.



7. A table of values of an increasing function f is shown. Use the table to find lower and upper estimates for $\int_{10}^{30} f(x)\,dx$.

x	10	14	18	22	26	30
$f(x)$	-12	-6	-2	1	3	8

8. The table gives the values of a function obtained from an experiment. Use them to estimate $\int_{3}^{9} f(x)\,dx$ using three equal subintervals with (a) right endpoints, (b) left endpoints, and (c) midpoints. If the function is known to be an increasing function, can you say whether your estimates are less than or greater than the exact value of the integral?

x	3	4	5	6	7	8	9
$f(x)$	-3.4	-2.1	-0.6	0.3	0.9	1.4	1.8

9–12 Use the Midpoint Rule with the given value of n to approximate the integral. Round the answer to four decimal places.

9. $\int_{2}^{10} \sqrt{x^3 + 1}\,dx, \quad n = 4$

10. $\int_{0}^{\pi/2} \cos^4 x\,dx, \quad n = 4$

11. $\int_{0}^{1} \sin(x^2)\,dx, \quad n = 5$

12. $\int_{1}^{5} x^2 e^{-x}\,dx, \quad n = 4$

13. Drug pharmacokinetics During testing of a new drug, researchers measured the plasma drug concentration of each test subject at 10-minute intervals. The average concentrations $C(t)$ are shown in the table, where t is measured in minutes and C is measured in μg/mL. Use the Midpoint Rule to estimate the integral $\int_{0}^{100} C(t)\,dt$. State the units.

t	0	10	20	30	40	50
$C(t)$	0	1.3	1.8	2.2	2.4	2.5

t	60	70	80	90	100
$C(t)$	2.4	2.3	2.0	1.6	1.1

14. Salicylic acid pharmacokinetics In the study cited in Example 5, the metabolite salicylic acid (SA) was rapidly formed and peak SA levels of about 4.2 μg/mL were reached after an hour. The concentration of SA was modeled by the function

$$C(t) = 11.4te^{-t}$$

where t is measured in hours and C is measured in μg/mL. Use the Midpoint Rule with eight subintervals to estimate the integral $\int_{0}^{4} C(t)\,dt$. State the units.

15–18 Express the limit as a definite integral on the given interval.

15. $\displaystyle\lim_{n\to\infty} \sum_{i=1}^{n} x_i \ln(1 + x_i^2)\,\Delta x, \quad [2, 6]$

16. $\displaystyle\lim_{n\to\infty} \sum_{i=1}^{n} \frac{\cos x_i}{x_i}\,\Delta x, \quad [\pi, 2\pi]$

17. $\displaystyle\lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{2x_i^* + (x_i^*)^2}\,\Delta x, \quad [1, 8]$

18. $\displaystyle\lim_{n\to\infty} \sum_{i=1}^{n} [4 - 3(x_i^*)^2 + 6(x_i^*)^5]\,\Delta x, \quad [0, 2]$

19–23 Use the form of the definition of the integral given in Theorem 4 to evaluate the integral.

19. $\int_{-1}^{5} (1 + 3x)\,dx$

20. $\int_{1}^{4} (x^2 + 2x - 5)\,dx$

21. $\int_{0}^{2} (2 - x^2)\,dx$

22. $\int_{0}^{5} (1 + 2x^3)\,dx$

23. $\int_{1}^{2} x^3\,dx$

24. (a) Find an approximation to the integral $\int_{0}^{4} (x^2 - 3x)\,dx$ using a Riemann sum with right endpoints and $n = 8$.
(b) Draw a diagram like Figure 3 to illustrate the approximation in part (a).
(c) Use Theorem 4 to evaluate $\int_{0}^{4} (x^2 - 3x)\,dx$.
(d) Interpret the integral in part (c) as a difference of areas and illustrate with a diagram like Figure 4.

25–26 Express the integral as a limit of Riemann sums. Do not evaluate the limit.

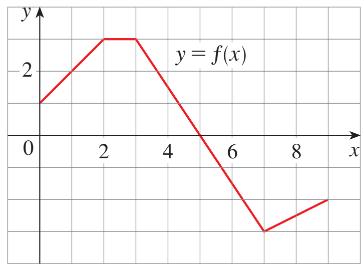
25. $\displaystyle\int_{2}^{6} \frac{x}{1 + x^5}\,dx$

26. $\int_{1}^{10} (x - 4\ln x)\,dx$

27. The graph of f is shown. Evaluate each integral by interpreting it in terms of areas.

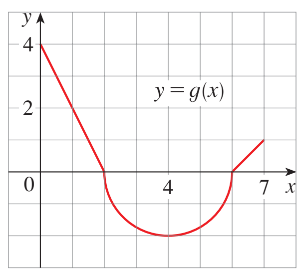
(a) $\int_{0}^{2} f(x)\,dx$ (b) $\int_{0}^{5} f(x)\,dx$

(c) $\int_{5}^{7} f(x)\,dx$ (d) $\int_{0}^{9} f(x)\,dx$



28. The graph of g consists of two straight lines and a semicircle. Use it to evaluate each integral.

(a) $\int_0^2 g(x)\,dx$ (b) $\int_2^6 g(x)\,dx$ (c) $\int_0^7 g(x)\,dx$



29–34 Evaluate the integral by interpreting it in terms of areas.

29. $\int_0^3 \left(\frac{1}{2}x - 1\right) dx$

30. $\int_{-2}^2 \sqrt{4 - x^2}\,dx$

31. $\int_{-3}^0 \left(1 + \sqrt{9 - x^2}\right) dx$

32. $\int_{-1}^3 (3 - 2x)\,dx$

33. $\int_{-1}^2 |x|\,dx$

34. $\int_0^{10} |x - 5|\,dx$

35. Evaluate $\int_\pi^\pi \sin^2 x \cos^4 x\,dx$.

36. Given that $\int_0^1 3x\sqrt{x^2 + 4}\,dx = 5\sqrt{5} - 8$, what is $\int_1^0 3u\sqrt{u^2 + 4}\,du$?

37. Write as a single integral in the form $\int_a^b f(x)\,dx$:

$$\int_{-2}^2 f(x)\,dx + \int_2^5 f(x)\,dx - \int_{-2}^{-1} f(x)\,dx$$

38. If $\int_1^5 f(x)\,dx = 12$ and $\int_4^5 f(x)\,dx = 3.6$, find $\int_1^4 f(x)\,dx$.

39. If $\int_0^9 f(x)\,dx = 37$ and $\int_0^9 g(x)\,dx = 16$, find $\int_0^9 [2f(x) + 3g(x)]\,dx$.

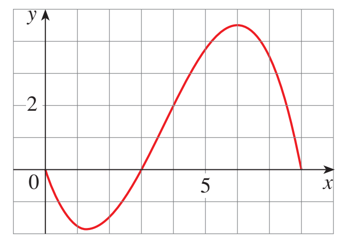
40. Find $\int_0^5 f(x)\,dx$ if

$$f(x) = \begin{cases} 3 & \text{for } x < 3 \\ x & \text{for } x \geqslant 3 \end{cases}$$

41. For the function f whose graph is shown, list the following quantities in increasing order, from smallest to largest, and explain your reasoning.

(A) $\int_0^8 f(x)\,dx$ (B) $\int_0^3 f(x)\,dx$ (C) $\int_3^8 f(x)\,dx$

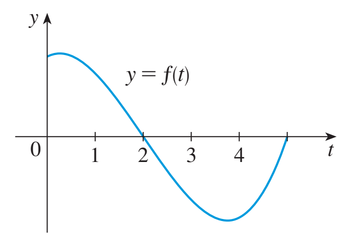
(D) $\int_4^8 f(x)\,dx$ (E) $f'(1)$



42. If $F(x) = \int_2^x f(t)\,dt$, where f is the function whose graph is given, which of the following values is largest?

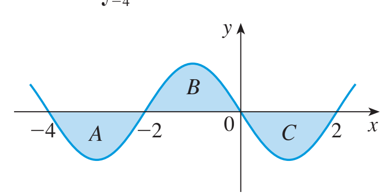
(A) $F(0)$ (B) $F(1)$ (C) $F(2)$

(D) $F(3)$ (E) $F(4)$



43. Each of the regions A, B, and C bounded by the graph of f and the x-axis has area 3. Find the value of

$$\int_{-4}^2 [f(x) + 2x + 5]\,dx$$



44. Suppose f has absolute minimum value m and absolute maximum value M. Between what two values must $\int_0^2 f(x)\,dx$ lie? Which property of integrals allows you to make your conclusion?

45. Use the properties of integrals to verify that

$$2 \leqslant \int_{-1}^1 \sqrt{1 + x^2}\,dx \leqslant 2\sqrt{2}$$

46. Use Property 8 to estimate the value of the integral

$$\int_0^2 \frac{1}{1 + x^2}\,dx$$

47–48 Express the limit as a definite integral.

47. $\lim_{n \to \infty} \sum_{i=1}^{n} \frac{i^4}{n^5}$ [*Hint:* Consider $f(x) = x^4$.]

48. $\lim_{n \to \infty} \frac{1}{n} \sum_{i=1}^{n} \frac{1}{1 + (i/n)^2}$

5.3 The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus is appropriately named because it establishes a connection between the two branches of calculus: differential calculus and integral calculus. Differential calculus arose from the tangent problem, whereas integral calculus arose from a seemingly unrelated problem, the area problem. Newton's mentor at Cambridge, Isaac Barrow (1630–1677), discovered that these two problems are actually closely related. In fact, he realized that differentiation and integration are inverse processes. The Fundamental Theorem of Calculus gives the precise inverse relationship between the derivative and the integral. It was Newton and Leibniz who exploited this relationship and used it to develop calculus into a systematic mathematical method.

Evaluating Definite Integrals

In Section 5.2 we computed integrals from the definition as a limit of Riemann sums and we saw that this procedure is sometimes long and difficult. Newton and Leibniz realized that they could calculate $\int_a^b f(x)\,dx$ if they happened to know an antiderivative F of f. Their discovery, called the Evaluation Theorem, is part of the Fundamental Theorem of Calculus, which is discussed later in this section.

Evaluation Theorem If f is continuous on the interval $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

where F is any antiderivative of f, that is, $F' = f$.

This theorem states that if we know an antiderivative F of f, then we can evaluate $\int_a^b f(x)\,dx$ simply by subtracting the values of F at the endpoints of the interval $[a, b]$. It is very surprising that $\int_a^b f(x)\,dx$, which was defined by a complicated procedure involving all of the values of $f(x)$ for $a \leqslant x \leqslant b$, can be found by knowing the values of $F(x)$ at only two points, a and b.

For instance, we know from Section 4.6 that an antiderivative of the function $f(x) = x^2$ is $F(x) = \frac{1}{3}x^3$, so the Evaluation Theorem tells us that

$$\int_0^1 x^2\,dx = F(1) - F(0) = \tfrac{1}{3} \cdot 1^3 - \tfrac{1}{3} \cdot 0^3 = \tfrac{1}{3}$$

Comparing this method with the calculation in Example 5.1.2, where we found the area under the parabola $y = x^2$ from 0 to 1 by computing a limit of sums, we see that the Evaluation Theorem provides us with a simple and powerful method.

Although the Evaluation Theorem may be surprising at first glance, it becomes plausible if we interpret it in physical terms. If $v(t)$ is the velocity of an object and $s(t)$ is its position at time t, then $v(t) = s'(t)$, so s is an antiderivative of v. In Section 5.1 we considered an object that always moves in the positive direction and made the guess that the area under the velocity curve is equal to the distance traveled. In symbols:

$$\int_a^b v(t)\,dt = s(b) - s(a)$$

That is exactly what the Evaluation Theorem says in this context.

PROOF OF THE EVALUATION THEOREM We divide the interval $[a, b]$ into n subintervals with endpoints $x_0 (= a), x_1, x_2, \ldots, x_n (= b)$ and with length $\Delta x = (b - a)/n$. Let F be any antiderivative of f. By subtracting and adding like terms, we can express the total difference in the F values as the sum of the differences over the subintervals:

$$\begin{aligned} F(b) - F(a) &= F(x_n) - F(x_0) \\ &= F(x_n) - F(x_{n-1}) + F(x_{n-1}) - F(x_{n-2}) + \cdots + F(x_2) - F(x_1) + F(x_1) - F(x_0) \\ &= \sum_{i=1}^{n} [F(x_i) - F(x_{i-1})] \end{aligned}$$

The Mean Value Theorem was discussed in Section 4.2.

Now F is continuous (because it's differentiable) and so we can apply the Mean Value Theorem to F on each subinterval $[x_{i-1}, x_i]$. Thus there exists a number x_i^* between x_{i-1} and x_i such that

$$F(x_i) - F(x_{i-1}) = F'(x_i^*)(x_i - x_{i-1}) = f(x_i^*)\,\Delta x$$

Therefore
$$F(b) - F(a) = \sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

Now we take the limit of each side of this equation as $n \to \infty$. The left side is a constant and the right side is a Riemann sum for the function f, so

$$F(b) - F(a) = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x = \int_a^b f(x)\,dx$$ ■

When applying the Evaluation Theorem we use the notation

$$F(x)\Big]_a^b = F(b) - F(a)$$

and so we can write

$$\int_a^b f(x)\,dx = F(x)\Big]_a^b \qquad \text{where} \qquad F' = f$$

Other common notations are $F(x)\big|_a^b$ and $[F(x)]_a^b$.

EXAMPLE 1 | Evaluate $\int_1^3 e^x\,dx$.

SOLUTION An antiderivative of $f(x) = e^x$ is $F(x) = e^x$, so we use the Evaluation Theorem as follows:

In applying the Evaluation Theorem we use a particular antiderivative F of f. It is not necessary to use the most general antiderivative $(e^x + C)$.

$$\int_1^3 e^x\,dx = e^x\Big]_1^3 = e^3 - e$$ ■

You can see from Example 1 that it is quite easy to calculate $\int_1^3 e^x\,dx$ with the Evaluation Theorem. Without the Evaluation Theorem it would be very difficult to calculate the integral. In fact if we use Theorem 5.2.4 with $f(x) = e^x$, $a = 1$, and $b = 3$, we get a challenging limit:

$$\int_1^3 e^x\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x = \lim_{n\to\infty} \frac{2}{n} \sum_{i=1}^{n} e^{1+2i/n}$$

This limit can be evaluated but it isn't easy to do so. The method of Example 1 is *much* easier.

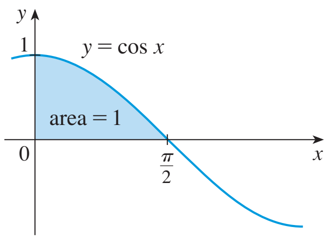


FIGURE 1

EXAMPLE 2 | Find the area under the cosine curve from 0 to b, where $0 \leqslant b \leqslant \pi/2$.

SOLUTION Since an antiderivative of $f(x) = \cos x$ is $F(x) = \sin x$, we have

$$A = \int_0^b \cos x\, dx = \sin x\Big]_0^b = \sin b - \sin 0 = \sin b$$

In particular, taking $b = \pi/2$, we have proved that the area under the cosine curve from 0 to $\pi/2$ is $\sin(\pi/2) = 1$. (See Figure 1.) ■

When the French mathematician Gilles de Roberval first found the area under the sine and cosine curves in 1635, this was a very challenging problem that required a great deal of ingenuity. If we didn't have the benefit of the Evaluation Theorem, we would have to compute a difficult limit of sums using obscure trigonometric identities (or a computer algebra system as in Exercise 5.1.23). It was even more difficult for Roberval because the apparatus of limits had not been invented in 1635. But in the 1660s and 1670s, when the Evaluation Theorem was discovered by Newton and Leibniz, such problems became very easy, as you can see from Example 2.

Indefinite Integrals

We need a convenient notation for antiderivatives that makes them easy to work with. Because of the relation given by the Evaluation Theorem between antiderivatives and integrals, the notation $\int f(x)\, dx$ is traditionally used for an antiderivative of f and is called an **indefinite integral**. Thus

$$\boxed{\int f(x)\, dx = F(x) \qquad \text{means} \qquad F'(x) = f(x)}$$

You should distinguish carefully between definite and indefinite integrals. A definite integral $\int_a^b f(x)\, dx$ is a *number*, whereas an indefinite integral $\int f(x)\, dx$ is a *function* (or family of functions). The connection between them is given by the Evaluation Theorem: If f is continuous on $[a, b]$, then

$$\int_a^b f(x)\, dx = \int f(x)\, dx\Big]_a^b$$

Recall from Section 4.6 that if F is an antiderivative of f on an interval I, then the most general antiderivative of f on I is $F(x) + C$, where C is an arbitrary constant. For instance, the formula

$$\int \frac{1}{x}\, dx = \ln|x| + C$$

is valid (on any interval that doesn't contain 0) because $(d/dx) \ln|x| = 1/x$. So an indefinite integral $\int f(x)\, dx$ can represent either a particular antiderivative of f or an entire *family* of antiderivatives (one for each value of the constant C).

The effectiveness of the Evaluation Theorem depends on having a supply of antiderivatives of functions. We therefore restate the Table of Antidifferentiation Formulas from Section 4.6, together with a few others, in the notation of indefinite integrals. Any formula can be verified by differentiating the function on the right side and obtaining the integrand. For instance,

$$\int \sec^2 x\, dx = \tan x + C \qquad \text{because} \qquad \frac{d}{dx}(\tan x + C) = \sec^2 x$$

(1) Table of Indefinite Integrals

$$\int [f(x) + g(x)]\, dx = \int f(x)\, dx + \int g(x)\, dx \qquad \int cf(x)\, dx = c\int f(x)\, dx$$

$$\int x^n\, dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1) \qquad \int \frac{1}{x}\, dx = \ln|x| + C$$

$$\int e^x\, dx = e^x + C \qquad \int e^{kx}\, dx = \frac{1}{k}e^{kx} + C$$

$$\int a^x\, dx = \frac{a^x}{\ln a} + C \qquad \int \sin x\, dx = -\cos x + C$$

$$\int \cos x\, dx = \sin x + C \qquad \int \sec^2 x\, dx = \tan x + C$$

$$\int \csc^2 x\, dx = -\cot x + C \qquad \int \sec x \tan x\, dx = \sec x + C$$

$$\int \csc x \cot x\, dx = -\csc x + C \qquad \int \frac{1}{x^2+1}\, dx = \tan^{-1} x + C$$

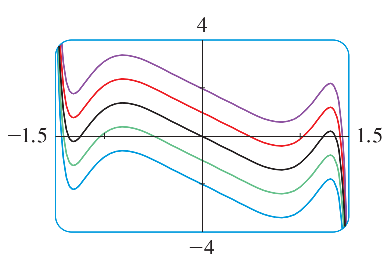


FIGURE 2

The indefinite integral in Example 3 is graphed in Figure 2 for several values of C. Here the value of C is the y-intercept.

EXAMPLE 3 | Find the general indefinite integral

$$\int (10x^4 - 2\sec^2 x)\, dx$$

SOLUTION Using our convention and Table 1 and properties of integrals, we have

$$\int (10x^4 - 2\sec^2 x)\, dx = 10\int x^4\, dx - 2\int \sec^2 x\, dx$$

$$= 10\,\frac{x^5}{5} - 2\tan x + C$$

$$= 2x^5 - 2\tan x + C$$

You should check this answer by differentiating it. ■

EXAMPLE 4 | Evaluate $\int_0^3 (x^3 - 6x)\, dx$.

SOLUTION Using the Evaluation theorem and Table 1, we have

$$\int_0^3 (x^3 - 6x)\, dx = \frac{x^4}{4} - 6\,\frac{x^2}{2}\Big]_0^3$$

$$= \left(\tfrac{1}{4}\cdot 3^4 - 3\cdot 3^2\right) - \left(\tfrac{1}{4}\cdot 0^4 - 3\cdot 0^2\right)$$

$$= \tfrac{81}{4} - 27 - 0 + 0 = -6.75$$

Compare this calculation with Example 5.2.2(b). ■

EXAMPLE 5 | Find $\int_0^2 \left(2x^3 - 6x + \frac{3}{x^2+1}\right) dx$ and interpret the result in terms of areas.

SOLUTION The Evaluation Theorem gives

$$\int_0^2 \left(2x^3 - 6x + \frac{3}{x^2+1}\right) dx = 2\frac{x^4}{4} - 6\frac{x^2}{2} + 3\tan^{-1}x\Big]_0^2$$

$$= \tfrac{1}{2}x^4 - 3x^2 + 3\tan^{-1}x\Big]_0^2$$

$$= \tfrac{1}{2}(2^4) - 3(2^2) + 3\tan^{-1}2 - 0$$

$$= -4 + 3\tan^{-1}2$$

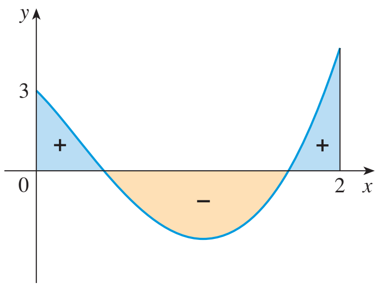


FIGURE 3

This is the exact value of the integral. If a decimal approximation is desired, we can use a calculator to approximate $\tan^{-1} 2$. Doing so, we get

$$\int_0^2 \left(2x^3 - 6x + \frac{3}{x^2+1}\right) dx \approx -0.67855$$

Figure 3 shows the graph of the integrand. We know from Section 5.2 that the value of the integral can be interpreted as a net area: the sum of the areas labeled with a plus sign minus the area labeled with a minus sign. ■

EXAMPLE 6 | Evaluate $\int_1^9 \frac{2t^2 + t^2\sqrt{t} - 1}{t^2}\, dt$.

SOLUTION First we need to write the integrand in a simpler form by carrying out the division:

$$\int_1^9 \frac{2t^2 + t^2\sqrt{t} - 1}{t^2}\, dt = \int_1^9 (2 + t^{1/2} - t^{-2})\, dt$$

$$= 2t + \frac{t^{3/2}}{\frac{3}{2}} - \frac{t^{-1}}{-1}\Big]_1^9 = 2t + \tfrac{2}{3}t^{3/2} + \frac{1}{t}\Big]_1^9$$

$$= \left(2 \cdot 9 + \tfrac{2}{3} \cdot 9^{3/2} + \tfrac{1}{9}\right) - \left(2 \cdot 1 + \tfrac{2}{3} \cdot 1^{3/2} + \tfrac{1}{1}\right)$$

$$= 18 + 18 + \tfrac{1}{9} - 2 - \tfrac{2}{3} - 1 = 32\tfrac{4}{9}$$ ■

The Net Change Theorem

The Evaluation Theorem says that if f is continuous on $[a, b]$, then

$$\int_a^b f(x)\, dx = F(b) - F(a)$$

where F is any antiderivative of f. This means that $F' = f$, so the equation can be rewritten as

$$\int_a^b F'(x)\, dx = F(b) - F(a)$$

We know that $F'(x)$ represents the rate of change of $y = F(x)$ with respect to x and $F(b) - F(a)$ is the change in y when x changes from a to b. [Note that y could, for instance, increase, then decrease, then increase again. Although y might change in both directions, $F(b) - F(a)$ represents the net change in y.] So we can reformulate the Evaluation Theorem in words as follows.

Net Change Theorem The integral of a rate of change is the net change:

$$\int_a^b F'(x)\,dx = F(b) - F(a)$$

EXAMPLE 7 | **Integrating rate of growth** If $N(t)$ is the size of a population at time t, explain the biological meaning of

$$\int_{t_1}^{t_2} \frac{dN}{dt}\,dt$$

SOLUTION The derivative dN/dt is the rate of growth of the population. According to the Net Change Theorem, we have

$$\int_{t_1}^{t_2} \frac{dN}{dt}\,dt = N(t_2) - N(t_1)$$

This is the net change in population during the time period from t_1 to t_2. The population increases when births happen and decreases when deaths occur. The net change takes into account both births and deaths. ■

EXAMPLE 8 | If an object moves along a straight line with position function $s(t)$, then its velocity is $v(t) = s'(t)$, so the Net Change Theorem says, in this context, that

$$\int_{t_1}^{t_2} v(t)\,dt = s(t_2) - s(t_1)$$

This is the net change of position, or *displacement*, of the particle during the time period from t_1 to t_2. In Section 5.1 we guessed that this was true for the case where the object moves in the positive direction, but now we have proved that it is always true. ■

■ The Fundamental Theorem

The first part of the Fundamental Theorem deals with functions defined by an equation of the form

(2) $$g(x) = \int_a^x f(t)\,dt$$

where f is a continuous function on $[a, b]$ and x varies between a and b. Observe that g depends only on x, which appears as the variable upper limit in the integral. If x is a fixed number, then the integral $\int_a^x f(t)\,dt$ is a definite number. If we then let x vary, the number $\int_a^x f(t)\,dt$ also varies and defines a function of x denoted by $g(x)$.

If f happens to be a positive function, then $g(x)$ can be interpreted as the area under the graph of f from a to x, where x can vary from a to b. (Think of g as the "area so far" function; see Figure 4.)

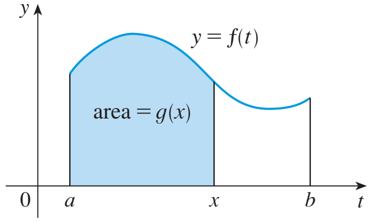


FIGURE 4

EXAMPLE 9 | If $g(x) = \int_a^x f(t)\,dt$, where $a = 1$ and $f(t) = t^2$, find a formula for $g(x)$ and calculate $g'(x)$.

SOLUTION In this case we can compute $g(x)$ explicitly using the Evaluation Theorem:

$$g(x) = \int_1^x t^2\,dt = \frac{t^3}{3}\bigg]_1^x = \frac{x^3 - 1}{3}$$

Then $$g'(x) = \frac{d}{dx}\left(\tfrac{1}{3}x^3 - \tfrac{1}{3}\right) = x^2$$ ■

For the function in Example 9, notice that $g'(x) = x^2$, that is, $g' = f$. In other words, if g is defined as the integral of f by Equation 2, then g turns out to be an antiderivative of f, at least in this case. To see why this might be generally true we consider any continuous function f with $f(x) \geqslant 0$. Then $g(x) = \int_a^x f(t)\,dt$ can be interpreted as the area under the graph of f from a to x, as in Figure 4.

In order to compute $g'(x)$ from the definition of a derivative we first observe that, for $h > 0$, $g(x + h) - g(x)$ is obtained by subtracting areas, so it is the area under the graph of f from x to $x + h$ (the blue area in Figure 5). For small h you can see from the figure that this area is approximately equal to the area of the rectangle with height $f(x)$ and width h:

$$g(x + h) - g(x) \approx hf(x)$$

so $$\frac{g(x + h) - g(x)}{h} \approx f(x)$$

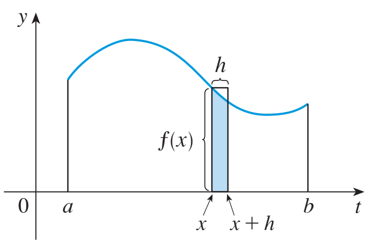


FIGURE 5

Intuitively, we therefore expect that

$$g'(x) = \lim_{h \to 0} \frac{g(x + h) - g(x)}{h} = f(x)$$

The fact that this is true, even when f is not necessarily positive, is the first part of the Fundamental Theorem of Calculus.

We abbreviate the name of this theorem as FTC1. In words, it says that the derivative of a definite integral with respect to its upper limit is the integrand evaluated at the upper limit.

The Fundamental Theorem of Calculus, Part 1 If f is continuous on $[a, b]$, then the function g defined by

$$g(x) = \int_a^x f(t)\,dt \qquad a \leqslant x \leqslant b$$

is an antiderivative of f, that is, $g'(x) = f(x)$ for $a < x < b$.

Using Leibniz notation for derivatives, we can write this theorem as

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$$

when f is continuous. Roughly speaking, this equation says that if we first integrate f and then differentiate the result, we get back to the original function f.

TEC Module 5.3 provides visual evidence for FTC1.

It is easy to prove the Fundamental Theorem if we make the assumption that f possesses an antiderivative F. (This is certainly plausible.) Then, by the Evaluation Theorem,

$$\int_a^x f(t)\,dt = F(x) - F(a)$$

for any x between a and b. Therefore

$$\frac{d}{dx}\int_a^x f(t)\,dt = \frac{d}{dx}[F(x) - F(a)] = F'(x) = f(x)$$

as required.

The amount of infection was defined on page 326.

EXAMPLE 10 | Measles pathogenesis In Section 5.1 we saw that the amount of infection exposed to the immune system by day t of a measles infection is

$$A(t) = \int_0^t f(s)\,ds$$

where $f(s) = -s(s - 21)(s + 1)$. What is the rate of change of the total amount of infection $A(t)$ at time t?

SOLUTION Since f is continuous, Part 1 of the Fundamental Theorem of Calculus implies that the rate of change of $A(t)$ at time t is

$$A'(t) = \frac{d}{dt}\int_0^t f(s)\,ds = f(t) = -t(t - 21)(t + 1)$$

EXAMPLE 11 | Find $\displaystyle\frac{d}{dx}\int_1^{x^4} \sec t\,dt$.

SOLUTION Here we have to be careful to use the Chain Rule in conjunction with Part 1 of the Fundamental Theorem. Let $u = x^4$. Then

$$\begin{aligned}\frac{d}{dx}\int_1^{x^4} \sec t\,dt &= \frac{d}{dx}\int_1^{u} \sec t\,dt \\ &= \frac{d}{du}\left[\int_1^{u} \sec t\,dt\right]\frac{du}{dx} && \text{(by the Chain Rule)} \\ &= \sec u\,\frac{du}{dx} && \text{(by FTC1)} \\ &= \sec(x^4)\cdot 4x^3\end{aligned}$$

Differentiation and Integration as Inverse Processes

We now bring together the two parts of the Fundamental Theorem. We regard Part 1 as fundamental because it relates integration and differentiation. But the Evaluation Theorem also relates integrals and derivatives, so we rename it as Part 2 of the Fundamental Theorem.

The Fundamental Theorem of Calculus Suppose f is continuous on $[a, b]$.

1. If $g(x) = \int_a^x f(t)\,dt$, then $g'(x) = f(x)$.

2. $\int_a^b f(x)\,dx = F(b) - F(a)$, where F is any antiderivative of f, that is, $F' = f$.

We noted that Part 1 can be rewritten as

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$$